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APYRASE INHIBITORS

Abstract

Provided are methods of using molecules as described herein in supporting crop viability and yield, such as by protecting crops from pests. In some cases, the methods inhibit apyrase enzymes by contacting such enzymes with a compound of the formula

##STR00001##

wherein G, Z, and R^{sup.1} are as described herein.

In further embodiments, an apyrase inhibitor as described herein is used in combination with one or more pesticide to treat a crop at risk of disease.

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Background/Summary

CROSS REFERENCE [0001] This application claims the benefit of the earlier filing date of U.S. Provisional Application No. 63/559,786, filed Feb. 29, 2024, the disclosure of which is incorporated herein by reference.

FIELD

[0002] The present disclosure relates to inhibitors of apyrase and methods for their use, in particular in the treatment of crops susceptible to pathogens.

INTRODUCTION

[0003] Crops are plagued worldwide by a variety of pests. These pests, such as insects, mites, nematodes, weeds and fungi, have developed an array of mechanisms for surviving pesticides, such as by sequestering, exporting or detoxifying them. The present inventors have discovered molecules and methods for potentiating the efficacy of pesticides by blocking certain mechanisms of resistance.

SUMMARY

[0004] Provided are methods of using molecules as described herein in supporting crop viability and yield, such as by protecting crops from pests. In some cases, the methods inhibit apyrase enzymes by contacting such enzymes with a compound of the formula

##STR00002##

wherein G, Z, and R^{sup.1} are as described herein.

[0005] In further embodiments, an apyrase inhibitor as described herein is used in combination with one or more pesticide to treat a crop at risk of disease.

Description

DETAILED DESCRIPTION

[0006] Before the present invention is described in greater detail, it is to be understood that this invention is not limited to particular embodiments described, as such may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting, since the scope of the present invention will be limited only by the appended claims.

[0007] Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limits of that range is also specifically disclosed. Each smaller range between any stated value or intervening value in a stated range and any other stated or intervening value in that stated range is encompassed within the invention. The upper and lower limits of these smaller ranges may independently be included or excluded in the range, and each range where either, neither or both limits are included in the smaller ranges is also encompassed within the invention, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the invention.

[0008] Unless defined otherwise, all technical and scientific terms used herein have the same

meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although any methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, some potential and exemplary methods and materials may now be described. Any and all publications mentioned herein are incorporated herein by reference to disclose and describe the methods and/or materials in connection with which the publications are cited. It is understood that the present disclosure supersedes any disclosure of an incorporated publication to the extent there is a contradiction.

[0009] It must be noted that as used herein and in the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise. It is further noted that the claims may be drafted to exclude any element, e.g., any optional element. As such, this statement is intended to serve as antecedent basis for use of such exclusive terminology as “solely”, “only” and the like in connection with the recitation of claim elements, or the use of a “negative” limitation.

[0010] The publications discussed herein are provided solely for their disclosure prior to the filing date of the present application. Further, the dates of publication provided may be different from the actual publication dates which may need to be independently confirmed. To the extent the definition or usage of any term herein conflicts with a definition or usage of a term in an application or reference incorporated by reference herein, the instant application shall control.

[0011] As will be apparent to those of skill in the art upon reading this disclosure, each of the individual embodiments described and illustrated herein has discrete components and features which may be readily separated from or combined with the features of any of the other several embodiments without departing from the scope or spirit of the present invention. Any recited method can be carried out in the order of events recited or in any other order which is logically possible.

Definitions

[0012] “Administering” refers to any suitable mode of administration, to control a fungal pathogen, including, treatment of an extant crop, seeds, soil or combination thereof.

[0013] “Control” with reference to a fungal pathogen, means block, inhibit and/or eradicate a fungal pathogen and/or prevent the fungal pathogen from damaging a crop. In one embodiment, control refers to the reduction of one or more fungi to undetectable levels, or to the reduction or suppression of a fungus to acceptable levels as determined by one of ordinary skill in the art (for example, a crop grower). Determinations of acceptable levels of fungus reduction are based on a number of factors, including to the crop, pathogen, severity of the pathogen, use restrictions, economic thresholds and other factors known to those of ordinary skill in the art.

[0014] As used herein, the terms “enhancer” and “potentiator”, refer to a compound or compounds disclosed herein that enhance the effects of a pesticide. Without limitation to theory the present enhancer compounds disclosed herein may function by blocking one or more pathways by which a pathogen, such as a fungal pathogen, evades toxicity, such as by detoxifying, sequestering or transporting a pesticide. In certain embodiment, the present compounds inhibit enzymatic apyrase activity which leads to the enhancement, accentuation or potentiation of a pesticide, such as an acaricide, antimicrobial, fungicide, herbicide, insecticide, molluscicide and/or nematocide. For example, when the enhancer or potentiator is used in conjunction with a fungicide, the combination of the potentiator and the fungicide enhances the fungicidal effect of the fungicide and/or renders a fungus that has become resistant to the fungicide susceptible to the fungicide as a result of the activity of the potentiator. Most often, these enhancers or potentiators do not themselves inhibit the growth of a fungus itself, nor do they have a detrimental effect on a living organism that is (or could be) infected with a fungus.

[0015] As used herein, the term “inoculation” refers to a method used to administer or apply an effective amount of a disclosed compound or formulation thereof to a target area of a field and/or plant. The inoculation method can be, but is not limited to, aerosol spray, pressure spray, direct

watering, and dipping. Target areas of a plant could include, but are not limited to, the leaves, roots, stems, buds, flowers, fruit, seed of the plant, and bulbs of the plant including bulb, corm, rhizoma, stem tuber, root tuber and rhizophore. Inoculation can include a method wherein a plant is treated in one area (for example, the root zone or foliage) and another area of the plant becomes protected (for example, foliage is inoculated when a disclosed compound is applied in the root zone or new growth when applied to foliage).

[0016] As used herein, the terms “wetable granule”, “water dispersible granule”, and “dispersible granule” refer to a solid granular formulation prepared by a granulation process, optionally containing fine particles of polymer-associated active ingredient, or aggregates of the same, a wetting agent and/or a dispersant, and optionally an inert filler. Wettable granules can be stored as a formulation, and can be provided to the market and/or end user without further processing. In some embodiments, they can be placed in a water-soluble bag for ease of use by the end user. In practical application, wettable granules are prepared for application by the end user. The wettable granules are mixed with water in the end user's spray tank to the proper dilution for the particular application. Dilution can vary by crop, target pathogen, time of year, geography, local regulations, and intensity of infection or pathogen load among other factors. Once properly diluted, the solution can be applied by spraying.

[0017] As used herein, the terms “wetable powder”, “water dispersible powder”, and “dispersible powder”, refer to a solid powdered formulation that contains active ingredient, optionally associated with a polymer, or aggregates of the same, and optionally one or more of a dispersant, a wetting agent, and an inert filler. Wettable powders can be stored as a formulation, and can be provided to the market and/or end user without further processing. In some embodiments, they can be placed in a water-soluble bag for ease of use by the end user. In practical application, a wettable powder is prepared for application by the end user. The wettable powder is mixed with water in the end user's spray tank to the proper dilution for the particular application. Dilution can vary by crop, fungal pathogen, time of year, geography, local regulations, and intensity of fungal load, among other factors. Once properly diluted, the solution can be applied by spraying.

[0018] As used herein, the term “high solids liquid suspension” refers to a liquid formulation that contains fine particles of active ingredient, or fine polymer particles associated with active ingredient, or aggregates of the same, a wetting agent and/or a dispersant, an anti-freezing agent, optionally an anti-settling agent or thickener, optionally a preservative, and water or oil as a carrier. High solids liquid suspensions can be stored as a formulation, and can be provided to the market and/or end user without further processing. In practical application, high solids liquid suspensions are prepared for application by the end user. The high solids liquid suspensions are mixed with water or oil in the end user's spray tank to the proper dilution for the particular application. Dilution can vary by crop, fungal pathogen, time of year, geography, local regulations, and intensity of infection among other factors. Once properly diluted, the solution can be applied by spraying.

[0019] As used herein, the term “phytologically acceptable” refers to compositions, diluents, excipients, and/or carriers that are generally applicable for use with any part of a plant during any part of its life cycle, including but not limited to seeds, seedlings, plant cells, plants, or flowers. The compositions can be prepared according to procedures, methods and formulas that are known to those of skill in the agricultural arts. Following the teachings of the present disclosure the artist skilled in the agricultural and/or chemical arts can readily prepare a desired composition. Most commonly, the compounds disclosed herein can be formulated to be stored, and/or applied, as aqueous or non-aqueous suspensions or emulsions prepared neat or from concentrated formulations of the compositions. Alternatively the compounds disclosed herein can be formulated for use in aerosol-generating equipment for application to agricultural produce stored in sealed chambers—an application method known as fogging. Water-soluble, water-suspendable or emulsifiable formulations comprising the presently disclosed compounds can also be converted into or formulated as solids (for example, wettable powders), which can then be diluted into a final

formulation. In certain formulations, the compositions of the present disclosure can also be provided in growth media, such as in vitro media for growth of plant or other types of cells, in laboratory plant growth media, in soil, or for spraying on seeds, seedlings, roots, stems, stalks, leaves, flowers or the entire plant.

[0020] Compounds herein can include all stereoisomers, including E and Z isomers, enantiomers, diastereomers, mixtures, racemates, atropisomers, and tautomers thereof.

[0021] Non-limiting examples of optional substituents include hydroxyl groups, sulfhydryl groups, halogens, amino groups, nitro groups, nitroso groups, cyano groups, azido groups, sulfoxide groups, sulfone groups, sulfonamide groups, carboxyl groups, carboxaldehyde groups, imine groups, alkyl groups, halo-alkyl groups, alkenyl groups, haloalkenyl groups, alkynyl groups, halo-alkynyl groups, alkoxy groups, aryl groups, aryloxy groups, aralkyl groups, arylalkoxy groups, heterocyclylalkyl groups, heteroaryl groups, cycloalkyl groups, acyl groups, acyloxy groups, carbamate groups, amide groups, ureido groups, epoxy groups, and ester groups.

[0022] “Alkyl” refers to an optionally substituted straight-chain, or optionally substituted branched-chain saturated hydrocarbon. Non-limiting examples of alkyl groups include straight, branched, and cyclic alkyl and alkylene groups. An alkyl group can be, for example, a C.sub.1, C.sub.2, C.sub.3, C.sub.4, C.sub.5, C.sub.6, C.sub.7, C.sub.8, C.sub.9, C.sub.10, C.sub.11, C.sub.12, C.sub.13, C.sub.14, C.sub.15, C.sub.16, C.sub.17, C.sub.18, C.sub.19, C.sub.20, C.sub.21, C.sub.22, C.sub.23, C.sub.24, C.sub.25, C.sub.26, C.sub.27, C.sub.28, C.sub.29, C.sub.30, C.sub.31, C.sub.32, C.sub.33, C.sub.34, C.sub.35, C.sub.36, C.sub.37, C.sub.38, C.sub.39, C.sub.40, C.sub.41, C.sub.42, C.sub.43, C.sub.44, C.sub.45, C.sub.46, C.sub.47, C.sub.48, C.sub.49, or C.sub.50 group that is substituted or unsubstituted. In some cases alkyl refers to a group having from one to about ten carbon atoms, or from one to six carbon atoms, wherein an sp³-hybridized carbon of the alkyl residue is attached to the rest of the molecule by a single bond. Whenever it appears herein, a numerical range such as “C.sub.1-6 alkyl” means that the alkyl group consists of 1 carbon atom, 2 carbon atoms, 3 carbon atoms, 4 carbon atoms, 5 carbon atoms or 6 carbon atoms, although the present definition also covers the occurrence of the term “alkyl” where no numerical range is designated. In some embodiments, the alkyl is a C.sub.1-10 alkyl, a C.sub.1-9 alkyl, a C.sub.1-8 alkyl, a C.sub.1-7 alkyl, a C.sub.1-6 alkyl, a C.sub.1-5 alkyl, a C.sub.1-4 alkyl, a C.sub.1-3 alkyl, a C.sub.1-2 alkyl, or a C.sub.1-8 alkyl.

[0023] Examples of alkyl groups include, but are not limited to, methyl, ethyl, n-propyl, isopropyl, 2-methyl-1-propyl, 2-methyl-2-propyl, 2-methyl-1-butyl, 3-methyl-1-butyl, 2-methyl-3-butyl, 2,2-dimethyl-1-propyl, 2-methyl-1-pentyl, 3-methyl-1-pentyl, 4-methyl-1-pentyl, 2-methyl-2-pentyl, 3-methyl-2-pentyl, 4-methyl-2-pentyl, 2,2-dimethyl-1-butyl, 3,3-dimethyl-1-butyl, 2-ethyl-1-butyl, n-butyl, isobutyl, sec-butyl, t-butyl, n-pentyl, isopentyl, neopentyl, tert-amyl, and hexyl, and longer alkyl groups, such as heptyl, octyl, and the like.

[0024] Non-limiting examples of straight alkyl groups include methyl, ethyl, propyl, butyl, pentyl, hexyl, heptyl, octyl, nonyl, and decyl.

[0025] Branched alkyl groups include any straight alkyl group substituted with any number of alkyl groups. Non-limiting examples of branched alkyl groups include isopropyl, isobutyl, sec-butyl, and t-butyl.

[0026] Unless stated otherwise specifically in the specification, an alkyl group is optionally substituted, for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, the alkyl is optionally substituted with oxo, halogen, —CN, —CF₃, —OH, —OMe, —NH₂, or —NO₂. In some embodiments, the alkyl is optionally substituted with oxo, halogen, —CN, —CF₃, —OH, or —OMe. In some embodiments, the alkyl is optionally substituted with halogen. Non-limiting examples of substituted alkyl groups includes hydroxymethyl, chloromethyl, trifluoromethyl, aminomethyl, 1-chloroethyl, 2-hydroxyethyl, 1,2-difluoroethyl, and 3-carboxypropyl.

[0027] “Alkenyl” refers to an optionally substituted straight-chain, or optionally substituted branched-chain hydrocarbon having one or more carbon-carbon double-bonds. The olefin or olefins of an alkenyl group can be, for example, E, Z, cis, trans, terminal, or exo-methylene. An alkenyl group can be, for example, a C.sub.2, C.sub.3, C.sub.4, C.sub.5, C.sub.6, C.sub.7, C.sub.8, C.sub.9, C.sub.10, C.sub.11, C.sub.12, C.sub.13, C.sub.14, C.sub.15, C.sub.16, C.sub.17, C.sub.18, C.sub.19, C.sub.20, C.sub.21, C.sub.22, C.sub.23, C.sub.24, C.sub.25, C.sub.26, C.sub.27, C.sub.28, C.sub.29, C.sub.30, C.sub.31, C.sub.32, C.sub.33, C.sub.34, C.sub.35, C.sub.36, C.sub.37, C.sub.38, C.sub.39, C.sub.40, C.sub.41, C.sub.42, C.sub.43, C.sub.44, C.sub.45, C.sub.46, C.sub.47, C.sub.48, C.sub.49, or C.sub.50 group that is substituted or unsubstituted. Non-limiting examples of alkenyl and alkenylene groups include ethenyl, prop-1-en-1-yl, isopropenyl, but-1-en-4-yl; 2-chloroethenyl, 4-hydroxybuten-1-yl, 7-hydroxy-7-methyloct-4-en-2-yl, and 7-hydroxy-7-methyloct-3,5-dien-2-yl.

[0028] Whenever it appears herein, a numerical range such as “C.sub.2-C.sub.6 alkenyl” means that the alkenyl group may consist of 2 carbon atoms, 3 carbon atoms, 4 carbon atoms, 5 carbon atoms, or 6 carbon atoms, although the present definition also covers the occurrence of the term “alkenyl” where no numerical range is designated. In some embodiments, the alkenyl is a C.sub.2-C.sub.10 alkenyl, a C.sub.2-C.sub.9 alkenyl, a C.sub.2-C.sub.5 alkenyl, a C.sub.2-C.sub.7 alkenyl, a C.sub.2-C.sub.6 alkenyl, a C.sub.2-C.sub.5 alkenyl, a C.sub.2-C.sub.4 alkenyl, a C.sub.2-C.sub.3 alkenyl, or a C.sub.2 alkenyl. Unless stated otherwise specifically in the specification, an alkenyl group is optionally substituted, for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, an alkenyl is optionally substituted with oxo, halogen, —CN, —CF.sub.3, —OH, —OMe, —NH.sub.2, or —NO.sub.2. In some embodiments, an alkenyl is optionally substituted with oxo, halogen, —CN, —CF.sub.3, —OH, or —OMe. In some embodiments, the alkenyl is optionally substituted with halogen.

[0029] “Alkynyl” refers to an optionally substituted straight-chain or optionally substituted branched-chain hydrocarbon. The triple bond of an alkynyl group can be internal or terminal. An alkynyl or alkynylene group can be, for example, a C.sub.2, C.sub.3, C.sub.4, C.sub.5, C.sub.6, C.sub.7, C.sub.8, C.sub.9, C.sub.10, C.sub.11, C.sub.12, C.sub.13, C.sub.14, C.sub.15, C.sub.16, C.sub.17, C.sub.18, C.sub.19, C.sub.20, C.sub.21, C.sub.22, C.sub.23, C.sub.24, C.sub.25, C.sub.26, C.sub.27, C.sub.28, C.sub.29, C.sub.30, C.sub.31, C.sub.32, C.sub.33, C.sub.34, C.sub.35, C.sub.36, C.sub.37, C.sub.38, C.sub.39, C.sub.40, C.sub.41, C.sub.42, C.sub.43, C.sub.44, C.sub.45, C.sub.46, C.sub.47, C.sub.48, C.sub.49, or C.sub.50 group that is substituted or unsubstituted. Non-limiting examples of alkynyl groups include ethynyl, prop-2-yn-1-yl, prop-1-yn-1-yl, and 2-methyl-hex-4-yn-1-yl; 5-hydroxy-5-methylhex-3-yn-1-yl, 6-hydroxy-6-methylhept-3-yn-2-yl, and 5-hydroxy-5-ethylhept-3-yn-1-yl.

[0030] Whenever it appears herein, a numerical range such as “C.sub.2-C.sub.6 alkynyl” means that the alkynyl group may consist of 2 carbon atoms, 3 carbon atoms, 4 carbon atoms, 5 carbon atoms, or 6 carbon atoms, although the present definition also covers the occurrence of the term “alkynyl” where no numerical range is designated. In some embodiments, the alkynyl is a C.sub.2-C.sub.10 alkynyl, a C.sub.2-C.sub.9 alkynyl, a C.sub.2-C.sub.5 alkynyl, a C.sub.2-C.sub.7 alkynyl, a C.sub.2-C.sub.6 alkynyl, a C.sub.2-C.sub.5 alkynyl, a C.sub.2-C.sub.4 alkynyl, a C.sub.2-C.sub.3 alkynyl, or a C.sub.2 alkynyl. Unless stated otherwise specifically in the specification, an alkynyl group is optionally substituted, for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, an alkynyl is optionally substituted with oxo, halogen, —CN, —CF.sub.3, —OH, —OMe, —NH.sub.2, or —NO.sub.2. In some embodiments, an alkynyl is optionally substituted with oxo, halogen, —CN, —CF.sub.3, —OH, or —OMe. In some embodiments, the alkynyl is optionally substituted with halogen.

[0031] A haloalkyl group can be any alkyl group substituted with any number of halogen atoms, for

example, fluorine, chlorine, bromine, and iodine atoms. A halo-alkenyl group can be any alkenyl group substituted with any number of halogen atoms. A haloalkynyl group can be any alkynyl group substituted with any number of halogen atoms.

[0032] An alkoxy group can be, for example, an oxygen atom substituted with any alkyl, alkenyl, or alkynyl group. An ether or an ether group comprises an alkoxy group. Non-limiting examples of alkoxy groups include methoxy, ethoxy, propoxy, isopropoxy, and isobutoxy.

[0033] The term “acyl” refers to the groups $\text{HC(O)}\text{—}$, $\text{alkyl-C(O)}\text{—}$, $\text{cycloalkyl-C(O)}\text{—}$, $\text{cycloalkenyl-C(O)}\text{—}$, $\text{aryl-C(O)}\text{—}$, $\text{heteroaryl-C(O)}\text{—}$ and $\text{heterocyclyl-C(O)}\text{—}$ where alkyl, cycloalkyl, cycloalkenyl, aryl, heteroaryl, and heterocyclyl are as described herein. By way of example acyl groups include acetyl and benzoyl groups.

[0034] “Alkoxy” refers to a radical of the formula $\text{—OR}^{\text{sup.a}}$ where $\text{R}^{\text{sup.a}}$ is an alkyl radical as defined. Unless stated otherwise specifically in the specification, an alkoxy group may be optionally substituted, for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, an alkoxy is optionally substituted with oxo, halogen, —CN , —CF_3 , —OH , —OMe , —NH_2 , or —NO_2 . In some embodiments, an alkoxy is optionally substituted with oxo, halogen, —CN , —CF_3 , —OH , or —OMe . In some embodiments, the alkoxy is optionally substituted with halogen.

[0035] “Aminoalkyl” refers to an alkyl radical, as defined above, that is substituted by one or more amines. In some embodiments, the alkyl is substituted with one amine. In some embodiments, the alkyl is substituted with one, two, or three amines. Hydroxyalkyl include, for example, aminomethyl, aminoethyl, aminopropyl, aminobutyl, or aminopentyl. In some embodiments, the hydroxyalkyl is aminomethyl.

[0036] “Aryl” refers to a radical derived from a hydrocarbon ring system comprising hydrogen, 6 to 30 carbon atoms, and at least one aromatic ring. The aryl radical may be a monocyclic, bicyclic, tricyclic, or tetracyclic ring system, which may include fused (when fused with a cycloalkyl or heterocyclylalkyl ring, the aryl is bonded through an aromatic ring atom) or bridged ring systems. In some embodiments, the aryl is a 6- to 10-membered aryl. In some embodiments, the aryl is a 6-membered aryl. Aryl radicals include, but are not limited to, aryl radicals derived from the hydrocarbon ring systems of anthrylene, naphthylene, phenanthrylene, anthracene, azulene, benzene, chrysene, fluoranthene, fluorene, as-indacene, s-indacene, indane, indene, naphthalene, phenalene, phenanthrene, pleiadene, pyrene, and triphenylene. In some embodiments, the aryl is phenyl. Unless stated otherwise specifically in the specification, an aryl may be optionally substituted, for example, with halogen, amino, nitrile, nitro, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, an aryl is optionally substituted with halogen, methyl, ethyl, —CN , —CF_3 , —OH , —OMe , —NH_2 , or —NO_2 . In some embodiments, an aryl is optionally substituted with halogen, methyl, ethyl, —CN , —CF_3 , —OH , or —OMe . In some embodiments, the aryl is optionally substituted with halogen.

[0037] “Cycloalkyl” refers to a stable, partially or fully saturated, monocyclic or polycyclic carbocyclic ring, which may include fused (when fused with an aryl or a heteroaryl ring, the cycloalkyl is bonded through a non-aromatic ring atom), bridged, or spiro ring systems. Representative cycloalkyls include, but are not limited to, cycloalkyls having from three to fifteen carbon atoms (C₃-C₁₅ cycloalkyl), from three to ten carbon atoms (C₃-C₁₀ cycloalkyl), from three to eight carbon atoms (C₃-C₈ cycloalkyl), from three to six carbon atoms (C₃-C₆ cycloalkyl), from three to five carbon atoms (C₃-C₅ cycloalkyl), or three to four carbon atoms (C₃-C₄ cycloalkyl). In some embodiments, the cycloalkyl is a 3- to 6-membered cycloalkyl. In some embodiments, the cycloalkyl is a 5- to 6-membered cycloalkyl. Non-limiting examples of cycloalkyl groups include cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, and cyclooctyl groups. Cycloalkyl groups also

include fused-, bridged-, and spiro-bicycles and higher fused-, bridged-, and spiro-systems. A cycloalkyl group can be substituted with any number of straight, branched, or cyclic alkyl groups. Non-limiting examples of cyclic alkyl groups include cyclopropyl, 2-methyl-cycloprop-1-yl, cycloprop-2-en-1-yl, cyclobutyl, 2,3-dihydroxycyclobut-1-yl, cyclobut-2-en-1-yl, cyclopentyl, cyclopent-2-en-1-yl, cyclopenta-2,4-dien-1-yl, cyclohexyl, cyclohex-2-en-1-yl, cycloheptyl, cyclooctanyl, 2,5-dimethylcyclopent-1-yl, 3,5-dichlorocyclohex-1-yl, 4-hydroxycyclohex-1-yl, 3,3,5-trimethylcyclohex-1-yl, octahydropentalenyl, octahydro-1H-indenyl, 3a,4,5,6,7,7a-hexahydro-3H-inden-4-yl, decahydroazulenyl, bicyclo-[2.1.1]hexanyl, bicyclo[2.2.1]heptanyl, bicyclo[3.1.1]heptanyl, 1,3-dimethyl[2.2.1]heptan-2-yl, bicyclo[2.2.2]octanyl, and bicyclo[3.3.3]undecanyl.

[0038] Monocyclic cycloalkyls include, for example, cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, and cyclooctyl.

[0039] Polycyclic cycloalkyls or carbocycles include, for example, adamantyl, norbornyl, decalinyl, bicyclo[3.3.0]octane, bicyclo[4.3.0]nonane, cis-decalin, trans-decalin, bicyclo[2.1.1]hexane, bicyclo[2.2.1]heptane, bicyclo[2.2.2]octane, bicyclo[3.2.2]nonane, and bicyclo[3.3.2]decane, and 7,7-dimethyl-bicyclo[2.2.1]heptanyl.

[0040] Partially saturated cycloalkyls include, for example, cyclopentenyl, cyclohexenyl, cycloheptenyl, and cyclooctenyl. Unless stated otherwise specifically in the specification, a cycloalkyl is optionally substituted, for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, a cycloalkyl is optionally substituted with oxo, halogen, methyl, ethyl, —CN, —CF₃, —OH, —OMe, —NH₂, or —NO₂. In some embodiments, a cycloalkyl is optionally substituted with oxo, halogen, methyl, ethyl, —CN, —CF₃, —OH, or —OMe. In some embodiments, the cycloalkyl is optionally substituted with halogen.

[0041] “Deuteroalkyl” refers to an alkyl radical, as defined above, that is substituted by one or more deuteriums. In some embodiments, the alkyl is substituted with one deuterium. In some embodiments, the alkyl is substituted with one, two, or three deuteriums. In some embodiments, the alkyl is substituted with one, two, three, four, five, or six deuteriums. Deuteroalkyl include, for example, CD₃, CH₂D, CHD₂, CH₂CD₂, CD₂CD₂, CHD₂CD₂, CH₂CH₂D₂, or CH₂CHD₂. In some embodiments, the deuteroalkyl is CD₃.

[0042] “Haloalkyl” refers to an alkyl radical, as defined above, that is substituted by one or more halogens. In some embodiments, the alkyl is substituted with one, two, or three halogens. In some embodiments, the alkyl is substituted with one, two, three, four, five, or six halogens. Haloalkyl include, for example, trifluoromethyl, difluoromethyl, fluoromethyl, trichloromethyl, 2,2,2-trifluoroethyl, 1,2-difluoroethyl, 3-bromo-2-fluoropropyl, 1,2-dibromoethyl, and the like. In some embodiments, the haloalkyl is trifluoromethyl.

[0043] “Halo” or “halogen” refers to bromo, chloro, fluoro, or iodo. In some embodiments, halogen is fluoro or chloro. In some embodiments, halogen is fluoro.

[0044] “Heteroalkyl” refers to an alkyl group in which one or more skeletal atoms of the alkyl are selected from an atom other than carbon, e.g., oxygen, nitrogen (e.g., —NH—, —N(alkyl)-), sulfur, or combinations thereof. A heteroalkyl is attached to the rest of the molecule at a carbon atom of the heteroalkyl. In one aspect, a heteroalkyl is a C₁₋₆ heteroalkyl wherein the heteroalkyl is comprised of 1 to 6 carbon atoms and one or more atoms other than carbon, e.g., oxygen, nitrogen (e.g., —NH—, —N(alkyl)-), sulfur, or combinations thereof wherein the heteroalkyl is attached to the rest of the molecule at a carbon atom of the heteroalkyl. Examples of such heteroalkyl are, for example, —CH₂OCH₃, —CH₂CH₂OCH₃, —CH₂CH₂CH₂OCH₃, or —CH(CH₃)OCH₃. Unless stated otherwise specifically in the specification, a heteroalkyl is optionally substituted for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, a heteroalkyl is

optionally substituted with oxo, halogen, methyl, ethyl, —CN, —CF.sub.3, —OH, —OMe, —NH.sub.2, or —NO.sub.2. In some embodiments, a heteroalkyl is optionally substituted with oxo, halogen, methyl, ethyl, —CN, —CF.sub.3, —OH, or —OMe. In some embodiments, the heteroalkyl is optionally substituted with halogen.

[0045] “Hydroxyalkyl” refers to an alkyl radical, as defined above, that is substituted by one or more hydroxyls. In some embodiments, the alkyl is substituted with one hydroxyl. In some embodiments, the alkyl is substituted with one, two, or three hydroxyls. Hydroxyalkyl include, for example, hydroxymethyl, hydroxyethyl, hydroxypropyl, hydroxybutyl, or hydroxypentyl. In some embodiments, the hydroxyalkyl is hydroxymethyl.

[0046] A heterocycle can be any ring containing a ring atom that is not carbon, for example, N, O, S, P, Si, B, or any other heteroatom. A heterocycle can be substituted with any number of substituents, for example, alkyl groups and halogen atoms. A heterocycle can be aromatic (heteroaryl) or non-aromatic. Non-limiting examples of heterocycles include pyrrole, pyrrolidine, pyridine, pyrimidine, pyrazine, pyridazine, piperidine, succinimide, maleimide, morpholine, imidazole, thiophene, furan, tetrahydrofuran, pyran, and tetrahydropyran.

[0047] “Heterocyclyl” refers to a stable 3- to 24-membered heterocycle. Non-limiting examples of heterocycles include: heterocyclic units having a single ring containing one or more heteroatoms, non-limiting examples of which include, diazirinyl, aziridinyl, azetidiny, pyrazolidinyl, imidazolidinyl, oxazolidinyl, isoxazolinyl, thiazolidinyl, isothiazolinyl, oxathiazolidinonyl, oxazolidinonyl, hydantoinyl, tetrahydrofuranyl, pyrrolidinyl, morpholinyl, piperazinyl, piperidinyl, dihydropyranyl, tetrahydropyranyl, piperidin-2-onyl, 2,3,4,5-tetrahydro-1H-azepinyl, 2,3-dihydro-1H-indole, and 1,2,3,4-tetrahydroquinoline; and ii) heterocyclic units having 2 or more rings one of which is a heterocyclic ring, non-limiting examples of which include hexahydro-1H-pyrroliziny, 3a,4,5,6,7,7a-hexahydro-1H-benzo[d]imidazolyl, 3a,4,5,6,7,7a-hexahydro-1H-indolyl, 1,2,3,4-tetrahydroquinolinyl, and decahydro-1H-cycloocta[b]pyrrollyl.

[0048] “Heterocyclylalkyl” or “heterocycloalkyl” or “cycloheteroalkyl” refers to a stable 3- to 24-membered partially or fully saturated ring radical comprising 2 to 23 carbon atoms and from one to 8 heteroatoms selected from the group consisting of nitrogen, oxygen, phosphorous, and sulfur. Unless stated otherwise specifically in the specification, the heterocyclylalkyl radical may be a monocyclic, bicyclic, tricyclic, or tetracyclic ring system, which may include fused (when fused with an aryl or a heteroaryl ring, the heterocyclylalkyl is bonded through a non-aromatic ring atom) or bridged ring systems; and the nitrogen, carbon, or sulfur atoms in the heterocyclylalkyl radical may be optionally oxidized; the nitrogen atom may be optionally quaternized.

[0049] Representative heterocyclylalkyls include, but are not limited to, heterocyclylalkyls having from two to fifteen carbon atoms (C.sub.2-C.sub.15 heterocyclylalkyl), from two to ten carbon atoms (C.sub.2-C.sub.10 heterocyclylalkyl), from two to eight carbon atoms (C.sub.2-C.sub.8 heterocyclylalkyl), from two to six carbon atoms (C.sub.2-C.sub.6 heterocyclylalkyl), from two to five carbon atoms (C.sub.2-C.sub.5 heterocyclylalkyl), or two to four carbon atoms (C.sub.2-C.sub.4 heterocyclylalkyl). In some embodiments, the heterocyclylalkyl is a 3- to 6-membered heterocyclylalkyl or a 3- to 8-membered heterocyclylalkyl. In some embodiments, the cycloalkyl is a 5- to 6-membered heterocyclylalkyl. Examples of such heterocyclylalkyl radicals include, but are not limited to, aziridinyl, azetidiny, dioxolanyl, thienyl[1,3]dithianyl, decahydroisoquinolyl, imidazolinyl, imidazolidinyl, isothiazolidinyl, isoxazolidinyl, morpholinyl, octahydroindolyl, octahydroisoindolyl, 2-oxopiperazinyl, 2-oxopiperidinyl, 2-oxopyrrolidinyl, oxazolidinyl, piperidinyl, piperazinyl, 4-piperidonyl, pyrrolidinyl, pyrazolidinyl, quinuclidinyl, thiazolidinyl, tetrahydrofuryl, trithianyl, tetrahydropyranyl, thiomorpholinyl, thiamorpholinyl, 1-oxo-thiomorpholinyl, 1,1-dioxo-thiomorpholinyl, 1,3-dihydroisobenzofuran-1-yl, 3-oxo-1,3-dihydroisobenzofuran-1-yl, methyl-2-oxo-1,3-dioxol-4-yl, and 2-oxo-1,3-dioxol-4-yl. The term heterocyclylalkyl also includes all ring forms of the carbohydrates, including but not limited to, the monosaccharides, the disaccharides, and the oligosaccharides. It is understood that when referring

to the number of carbon atoms in a heterocyclylalkyl, the number of carbon atoms in the heterocyclylalkyl is not the same as the total number of atoms (including the heteroatoms) that make up the heterocyclylalkyl (i.e. skeletal atoms of the heterocyclylalkyl ring). Unless stated otherwise specifically in the specification, a heterocyclylalkyl is optionally substituted, for example, with oxo, halogen, amino, nitrile, nitro, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, a heterocyclylalkyl is optionally substituted with oxo, halogen, methyl, ethyl, —CN, —CF₃, —OH, —OMe, —NH₂, or —NO₂. In some embodiments, a heterocyclylalkyl is optionally substituted with oxo, halogen, methyl, ethyl, —CN, —CF₃, —OH, or —OMe. In some embodiments, the heterocyclylalkyl is optionally substituted with halogen.

[0050] “Heteroaryl” refers to a 5- to 14-membered ring system radical comprising hydrogen atoms, one to thirteen carbon atoms, one to six heteroatoms selected from the group consisting of nitrogen, oxygen, phosphorous, and sulfur, and at least one aromatic ring. The heteroaryl radical may be a monocyclic, bicyclic, tricyclic, or tetracyclic ring system, which may include fused (when fused with a cycloalkyl or heterocyclylalkyl ring, the heteroaryl is bonded through an aromatic ring atom) or bridged ring systems; and the nitrogen, carbon, or sulfur atoms in the heteroaryl radical may be optionally oxidized; the nitrogen atom may be optionally quaternized. In some embodiments, the heteroaryl is a 5- to 10-membered heteroaryl. In some embodiments, the heteroaryl is a 5- to 6-membered heteroaryl. Examples include, but are not limited to, azepinyl, acridinyl, benzimidazolyl, benzothiazolyl, benzindolyl, benzodioxolyl, benzofuranyl, benzooxazolyl, benzothiazolyl, benzothiadiazolyl, benzo[b][1,4]dioxepinyl, 1,4-benzodioxanyl, benzonaphthofuranyl, benzoxazolyl, benzodioxolyl, benzodioxinyl, benzopyranyl, benzopyranonyl, benzofuranyl, benzofuranonyl, benzothienyl (benzothiophenyl), benzotriazolyl, benzo[4,6]imidazo[1,2-a]pyridinyl, carbazolyl, cinnolinyl, dibenzofuranyl, dibenzothiophenyl, furanyl, furanonyl, isothiazolyl, imidazolyl, indazolyl, indolyl, indazolyl, isoindolyl, indolinyl, isoindolinyl, isoquinolyl, indoliziny, isoxazolyl, naphthyridinyl, oxadiazolyl, 2-oxoazepinyl, oxazolyl, oxiranyl, 1-oxidopyridinyl, 1-oxidopyrimidinyl, 1-oxidopyrazinyl, 1-oxidopyridazinyl, 1-phenyl-1H-pyrrolyl, phenazinyl, phenothiazinyl, phenoxazinyl, phthalazinyl, pteridinyl, purinyl, pyrrolyl, pyrazolyl, pyridinyl, pyrazinyl, pyrimidinyl, pyridazinyl, quinazolinyl, quinoxalinyl, quinolinyl, quinuclidinyl, isoquinolinyl, tetrahydroquinolinyl, thiazolyl, thiadiazolyl, triazolyl, tetrazolyl, triazinyl, and thiophenyl (i.e., thienyl). Unless stated otherwise specifically in the specification, a heteroaryl is optionally substituted, for example, with halogen, amino, nitrile, nitro, hydroxyl, alkyl, alkenyl, alkynyl, haloalkyl, alkoxy, aryl, cycloalkyl, heterocyclylalkyl, heteroaryl, and the like. In some embodiments, a heteroaryl is optionally substituted with halogen, methyl, ethyl, —CN, —CF₃, —OH, —OMe, —NH₂, or —NO₂. In some embodiments, a heteroaryl is optionally substituted with halogen, methyl, ethyl, —CN, —CF₃, —OH, or —OMe. In some embodiments, the heteroaryl is optionally substituted with halogen.

Compounds

[0051] Provided are apyrase inhibitor compounds. In some cases, the compound has formula (I): ##STR00003## [0052] wherein: [0053] G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R^{sup.a} and R^{sup.b}; [0054] Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C₁₋₆ alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R^{sup.a} and R^{sup.b}; [0055] R^{sup.1} is hydrogen or C₁₋₆ alkyl; [0056] or Z and R^{sup.1} together with the nitrogen to which they are attached form a 5-, 6-, or 7-membered ring selected from heterocyclylalkyl rings optionally substituted with 1, 2 or 3 groups selected from R^{sup.a} and R^{sup.b}; [0057] each R^{sup.a} is independently selected from the group consisting of C₁₋₆ alkyl, C₃₋₈ cycloalkyl, C₅₋₁₀ aryl, C₆₋₁₀ arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclylalkyl, 4-11 membered heterocyclylalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl; [0058] each R^{sup.a} is optionally

substituted with one or more groups selected from R^{sup.b} and R^{sup.e}; [0059] each R^{sup.b} is independently selected from the group consisting of =O, —OR^{sup.d}, —OCF₃, —OCF₂H, —SR^{sup.d}, =NR^{sup.d}, =NOR^{sup.d}, —NR^{sup.c}R^{sup.c}, halogen, —CF₃, —CF₂H, —CN, —NO₂, =N₂, —N₃, —S(O)R^{sup.d}, —S(O)₂R^{sup.d}, —S(O)₂OR^{sup.d}, —S(O)NR^{sup.c}R^{sup.c}, —S(O)₂NR^{sup.c}R^{sup.c}, —NHS(O)₂R^{sup.d}, —OS(O)R^{sup.d}, —OS(O)₂R^{sup.d}, —OS(O)₂OR^{sup.d}, —OS(O)₂NR^{sup.c}R^{sup.c}, —C(O)R^{sup.d}, —C(O)OR^{sup.d}, —C(O)NR^{sup.c}R^{sup.c}, —C(NH)NR^{sup.c}R^{sup.c}, —C(NR^{sup.a})NR^{sup.c}R^{sup.c}, —C(NOH)R^{sup.a}, —C(NOH)NR^{sup.c}R^{sup.c}, —OC(O)R^{sup.d}, —OC(O)OR^{sup.d}, —OC(O)NR^{sup.c}R^{sup.c}, —OC(NH)NR^{sup.c}R^{sup.c}, —OC(NR^{sup.a})NR^{sup.c}R^{sup.c}, —[NHC(O)]_nR^{sup.d}, —[NR^{sup.c}(O)]_nR^{sup.d}, —[NHC(O)]_nOR^{sup.d}, —[NR^{sup.a}C(O)]_nOR^{sup.d}, —[NHC(O)]_nNR^{sup.c}R^{sup.c}, —[NR^{sup.a}C(O)]_nNR^{sup.c}R^{sup.c}, —NHSO₂R^{sup.d}, —[NHC(NH)]_nNR^{sup.c}R^{sup.c} and —[NR^{sup.a}C(NR^{sup.d})]_nNR^{sup.c}R^{sup.c}; [0060] each R^{sup.c} is independently hydrogen, R^{sup.a}, or, alternatively, two R^{sup.c} are taken together with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of the same or different group selected from R^{sup.a}, =O, halogen, and R^{sup.e}; [0061] each R^{sup.d} is independently C₁₋₆ alkyl or C₃₋₈ cycloalkyl; [0062] each R^{sup.e} is independently C₁₋₆ alkyl, C₃₋₈ cycloalkyl, —C(O)R^{sup.d}, and —C(O)OR^{sup.d}; [0063] provided that the compound does not have the formula

##STR00004##

Group G

[0064] In some embodiments, G comprises a six-membered aryl or six-membered heteroaryl ring, such as phenyl or pyridine. In some embodiments, G is optionally substituted phenyl, i.e. G can be unsubstituted phenyl or it can be phenyl substituted with R^{sup.a}, R^{sup.b}, and R^{sup.a} substituted with one or more of the same or different R^{sup.b}.

[0065] In some cases, G is substituted phenyl. In some cases, G is a phenyl group substituted at its para position, such as with a C₁₋₆ alkyl group. In some cases, G is a phenyl group substituted at its para position with a n-propyl group, i.e. G has formula (Ia):

##STR00005##

[0066] In some cases, G is a phenyl group substituted at its para position with an Ar group that is an optionally substituted aryl or heteroaryl group, as shown below in formula (Ib). As such, Ar in formula (Ib) can be an unsubstituted aryl group, a substituted aryl group, an unsubstituted heteroaryl group, or a substituted heteroaryl group.

##STR00006##

[0067] In some embodiments of formula (Ib), Ar is unsubstituted, substituted with an R^{sup.a} group of C₁₋₆ alkyl, or substituted with an R^{sup.b} group of halogen.

[0068] In some cases, G is an optionally substituted biphenyl group, i.e. G is a phenyl group that is substituted with an R^{sup.a} group of phenyl. In some cases, the compound has formula (Ib1), i.e. wherein G is an unsubstituted biphenyl group.

##STR00007##

[0069] In some cases, G is an optionally substituted pyrimidinyl. In some cases, the compound has formula (Ic), wherein the sulfonamide is at the 5-position of the pyrimidine ring, and there is an Ar group at the 2-position of the pyrimidine ring, wherein the Ar group of formula (Ic) is optionally substituted aryl or heteroaryl. If “Ar” is aryl, then it will be a C₅₋₁₀ aryl, since that is the option for R^{sup.a} recited in relation to formula (I). If “Ar” is heteroaryl, then it will be a 5-10 membered heteroaryl, since that is the option for R^{sup.a} recited in relation to formula (I).

##STR00008##

wherein Ar is optionally substituted aryl or heteroaryl.

[0070] In some cases, the Ar of formula (Ic) is unsubstituted, substituted with an R.sup.a group of C.sub.1-6 alkyl, or substituted with an R.sup.b group of halogen.

Group Z

[0071] As discussed above, Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C.sub.1-6 alkyl.

[0072] In some cases, Z is substituted phenyl. For instance, Z can be phenyl substituted with an optionally substituted aryl or heteroaryl group. In some cases, Z is phenyl substituted with an optionally substituted alkyl group. In some cases, Z is phenyl substituted with an R.sup.b group.

[0073] In some cases, the compound has formula (Id):

##STR00009##

[0074] In some cases, the compound has formula (Ie):

##STR00010##

[0075] In formulas (Id) and (Ie), the Z group is a phenyl group that is substituted with a sulfonamide (—NHSO.sub.2—) or (—NHSO.sub.2—) group, as allowed for in the description of the options for formula (I).

[0076] In some cases, Z is phenyl substituted with an R.sup.b group of halogen or —CF.sub.3.

[0077] In some cases, Z is optionally substituted monocyclic heteroaryl, such as pyridyl or pyrimidyl.

[0078] In certain cases, Z and R.sup.1 together with the nitrogen to which they are attached form a 5-, 6-, or 7-membered ring selected from heterocyclalkyl rings optionally substituted with 1, 2 or 3 groups selected from R.sup.a and R.sup.b. In one such embodiment wherein Z and R.sup.1 together form a 5-, 6-, or 7-membered ring, the ring formed by Z and R.sup.1 together contains one or more additional heteroatoms. In one aspect of such embodiments, the one or more additional heteroatoms are selected from the group consisting of oxygen, nitrogen and sulfur. In such embodiments wherein an additional heteroatom is nitrogen, the nitrogen atom optionally is substituted. Similarly, in such embodiments wherein an additional heteroatom is sulfur, the sulfur atom optionally is substituted.

[0079] In some cases, Z and R.sup.1 together with the nitrogen to which they are attached form a fused 5-membered or 6-membered ring selected from heterocyclalkyl optionally substituted with 1, 2 or 3 groups selected from R.sup.a and R.sup.b.

[0080] In one embodiment, wherein Z and R.sup.1 together form a 5-membered or 6-membered ring, the ring formed by Z and R.sup.1 together contains one or more additional heteroatoms. In one such embodiment, the ring formed by Z and R.sup.1 together is a morpholinyl or piperazinyl ring.

[0081] In one embodiment, wherein Z and R.sup.1 together form a 5-membered or 6-membered ring, the ring formed by Z and R.sup.1 together is fused to an aromatic or heteroaromatic ring, which may be substituted or unsubstituted. In one such embodiment wherein Z and R.sup.1 together form a 5-membered or 6-membered ring, the ring formed by Z and R.sup.1 together is fused to an aromatic or heteroaromatic ring, which is substituted with 1, 2 or 3 groups selected from R.sup.a and R.sup.b. In one embodiment wherein a ring formed by Z and R.sup.1 together is fused to an aromatic or heteroaromatic ring, the resulting fused ring system is an indoline ring system.

[0082] In some embodiments, Z is selected from the group consisting of:

##STR00011## ##STR00012##

[0083] In some cases, features of the G groups can be combined with features of the Z groups. For instance, in some cases the G group shown in formulas (Ia), (Ib), (Ib1), or (Ic) can be combined with the Z group shown in formulas (Id) or (Ie) in any possible combination.

Additional Aspects

[0084] As discussed above, each R.sup.d is independently C.sub.1-6 alkyl or C.sub.3-8 cycloalkyl. In some cases, R.sup.d is C.sub.1-3 alkyl or C.sub.3-4 cycloalkyl.

[0085] As discussed above, each R.sup.e is independently C.sub.1-6 alkyl or C.sub.3-8 cycloalkyl. In some cases, R.sup.e is C.sub.1-3 alkyl or C.sub.3-4 cycloalkyl.

[0086] As discussed above, many groups can be substituted with R.sup.a or R.sup.b. Additionally, R.sup.a can be substituted with R.sup.b. In some cases, R.sup.b is a group that includes an R.sup.c group, which can be R.sup.a. In some cases, R.sup.b is a group that includes a R.sup.d group, which is R.sup.a. There can be a limit on the number of times that a substituted is substituted with another substituent. For example, in some cases R.sup.a is not substituted with R.sup.b.

[0087] In some cases, the size of a group can be limited by specifying the number of non-hydrogen atoms in the group. For example, in some cases Z and its substituents have 6 to 20 non-hydrogen atoms, such as 6 to 16 non-hydrogen atoms or 6 to 12 non-hydrogen atoms. These numbers of non-hydrogen atoms include all levels of substitution. For example, in Compound A-1 shown below in the next section, Z can be interpreted as a phenyl (6 atoms) substituted with an R.sup.a of methyl (1 atom) and R.sup.a of tetrazole (5 atoms), wherein the tetrazole is further substituted with R.sup.b of methyl (1 atom), thereby giving a total non-hydrogen atom could of 6+1+5+1 or 13 atoms. Thus, the methyl on the tetrazole is a second-level substitution of the original Z group of phenyl, but this methyl on tetrazole is still counted towards the number of non-hydrogen atoms for Z.

[0088] Similarly, in some cases the G group can have a certain number of non-hydrogen atoms, such as 6 to 20 non-hydrogen atoms, 6 to 16 non-hydrogen atoms, or 6 to 12 non-hydrogen atoms. In Compound A-1 of the next section, G is a phenyl group (6 atoms) substituted with an R.sup.a group of methyl (3 atoms) for a total of 9 non-hydrogen atoms in G.

Specific Embodiments of Compounds

[0089] In some cases, the compound has a structure selected from Table 1, which is shown below.

TABLE-US-00001 TABLE 1 Compound Structure IUPAC Name A-1 [00013]

N-(4-methyl-3-(5-methyl-1H- tetrazol-1-yl)phenyl)-4- propylbenzenesulfonamide A-3 [00014]

N-(3-methoxyphenyl)-[1,1'- biphenyl]-4-sulfonamide A-4 [00015]

N-(2,5-dimethoxyphenyl)- [1,1'-biphenyl]-4-sulfonamide A-5 [00016]

N-(2,5-dimethylphenyl)-[1,1'- biphenyl]-4-sulfonamide A-6 [00017]

N-(2-(2-oxopyrrolidin-1- yl)phenyl)-4- propylbenzenesulfonamide A-7 [00018]

N-(3-(2-oxopyrrolidin-1- yl)phenyl)-4- propylbenzenesulfonamide A-8 [00019]

N-(2-oxoindolin-5-yl)-[1,1'- biphenyl]-4-sulfonamide A-9 [00020]

N-(o-tolyl)-[1,1'-biphenyl]-4- sulfonamide A-10 [00021]

N-methyl-N-phenyl-[1,1'- biphenyl]-4-sulfonamide A-11 [00022]

1-([1,1'- biphenyl]-4- ylsulfonyl)-4-(tetrahydro-2H- pyran-4-yl)piperazine A-12 [00023]

methyl 4-(N-(m- tolyl)sulfamoyl)benzoate A-13 [00024]

N-(4H-1,2,4-triazol-3-yl)- [1,1'-biphenyl]-4-sulfonamide A-14 [00025]

N-(3-(pyridin-3-yl)-1H- pyrazol-5-yl)-[1,1'-biphenyl]- 4-sulfonamide A-15 [00026]

N-(2-aminoethyl)- [1,1'- biphenyl]-4-sulfonamide hydrochloride A-16 [00027]

N-(2-fluoro-6- (hydroxymethyl)phenyl)-4- propylbenzenesulfonamide A-18 [00028]

4-(N-(m- tolyl)sulfamoyl)benzamide A-19 [00029]

2-methyl-3-(4-((4- propylphenyl)sulfonamido) phenyl)propanoic acid A-20 [00030]

N-(pyrimidin- 2-yl)-[1,1'- biphenyl]-4-sulfonamide A-21 [00031]

4-(cyanomethyl)-N-(4- methylpyrimidin-2- yl)benzenesulfonamide A-22 [00032]

N-(2-(3-methyl-4H- 1,2,4- triazol-4-yl)phenyl)-4- propylbenzenesulfonamide A-23 [00033]

N-(4- methyl-3-oxo-3,4- dihydropyrazin-2-yl)-[1,1'- biphenyl]-4-sulfonamide A-24 [00034]

N-(2-chloro-5-cyanopyridin- 3-yl)-[1,1'-biphenyl]-4- sulfonamide A-25

methyl 3-((4- propylphenyl)sulfonamido)-4- sulfamoylbenzoate A-26

2-hydroxy-5-((4- propylphenyl)sulfonamido) benzoic acid A-27

N-(2-((4-methyl-4H-1,2,4- triazol-3-yl)thio)phenyl)-4-

propylbenzenesulfonamide A-28 [00038]

N-(3-(1-cyclopropyl-1H- tetrazol-5-yl)phenyl)-4- propylbenzenesulfonamide A-29 [00039]

N-(4-(1,3,4-thiadiazol-

2-yl)phenyl)-[1,1'-biphenyl]-4- sulfonamide A-30 [00040]  N-(4-(1,1-dioxidoisothiazolidin-2-yl)phenyl)-4- propylbenzenesulfonamide A-31 [00041]  N-(4-((4- propylphenyl)sulfonamido) phenyl)cyclopropanecarbox- amide A-32 [00042]  N-(4-(ethylsulfonyl)phenyl)- 4-propylbenzenesulfonamide A-33 [00043]  1-([1,1'-biphenyl]-4- ylsulfonyl)piperidin-4-amine A-34 [00044]  N-(3-methoxyphenyl)-4'- methyl-[1,1'-biphenyl]-4- sulfonamide A-35 [00045]  N-(2-fluoro-5-((4- propylphenyl)sulfonamido) phenyl)acetamide A-36 [00046]  N-(benzo[d][1,3]dioxol-5- ylmethyl)-[1,1'-biphenyl]-4- sulfonamide A-37 [00047]  2-phenyl-N-(3- (trifluoromethyl)phenyl)pyrim- idine-5-sulfonamide A-38 [00048]  N-(2-hydroxy-5-((4- propylphenyl)sulfonamido) phenyl)acetamide A-39 [00049]  methyl 1-(2-((4- propylphenyl)sulfonamido) phenyl)-1H-1,2,3- triazole-4- carboxylate A-40 [00050]  N-(4-(2-methylpyrimidin-4- yl)phenyl)-4- propylbenzenesulfonamide A-41 [00051]  N-methyl-2-(2-((4- propylphenyl)sulfonamido) phenyl)acetamide A-42 [00052]  2-fluoro-6-((4- propylphenyl)sulfonamido) benzoic acid A-43 [00053]  5-fluoro-2-((4- propylphenyl)sulfonamido) benzoic acid A-44 [00054]  4-((4- propylphenyl)sulfonamido) benzoic acid A-45 [00055]  4,5-dimethoxy-2-((4- propylphenyl)sulfonamido) benzoic acid A-46 [00056]  2-(4-((4- propylphenyl)sulfonamido) phenyl)acetic acid A-47 [00057]  N-(5-chloro-2- morpholinophenyl)-4- propylbenzenesulfonamide A-48 [00058]  N-(2- ethylphenyl)-4- propylbenzenesulfonamide A-49 [00059]  N-(5-chloro-2,4- dimethoxyphenyl)-4- propylbenzenesulfonamide A-50 [00060]  N-(4- (methylsulfonyl)phenyl)-4- propylbenzenesulfonamide A-51 [00061]  N-(2- bromo-4,6- difluorophenyl)-4- propylbenzenesulfonamide A-52 [00062]  N-(azepan-2-ylidene)-4-((4- propylphenyl)sulfonamido) benzenesulfonamide A-53 [00063]  methyl 2-((4- propylphenyl)sulfonamido) benzoate A-54 [00064]  N-(2,3- dihydrobenzo[b][1,4]dioxin- 6-yl)-4- propylbenzenesulfonamide A-55 [00065]  N-(2-bromophenyl)-4- propylbenzenesulfonamide A-56 [00066]  4-chloro-N-(m- tolyl)benzenesulfonamide A-57 [00067]  N-(m- tolyl)benzenesulfonamide A-58 [00068]  4-fluoro-N-(m- tolyl)benzenesulfonamide A-59 [00069]  4-isopropyl-N-(m- tolyl)benzenesulfonamide A-60 [00070]  4-(pyrrolidin-1-ylsulfonyl)-N- (m- tolyl)benzenesulfonamide A-61 [00071]  3-(4-(N-(m- tolyl)sulfamoyl)phenyl)acrylic acid A-62 [00072]  4-propyl-N-(m- tolyl)benzenesulfonamide A-63 [00073]  N-(m-tolyl)-4- (trifluoromethoxy)benzenesul- fonamide A-64 [00074]  4-cyclohexyl-N-(m- tolyl)benzenesulfonamide A-65 [00075]  4-(N-(m- tolyl)sulfamoyl)benzoic acid A-66 [00076]  4-(N-(m-tolyl)sulfamoyl)-N- (2,2,2- trifluoroethyl)benzamide A-67 [00077]  N-(5-chloro-2-((4-methyl-4H- 1,2,4-triazol-3- yl)thio)phenyl)-4- propylbenzenesulfonamide A-68 [00078]  N-(4-chloro-2,5- dimethoxyphenyl)-4- propylbenzenesulfonamide A-69 [00079]  N-(1H-indazol-5-yl)-4- propylbenzenesulfonamide A-70 [00080]  N-(3-nitrophenyl)-4- propylbenzenesulfonamide A-71 [00081]  N-(2-chlorophenyl)-4- propylbenzenesulfonamide A-72 [00082]  4-propyl-N-(3- sulfamoylphenyl)benzene- sulfonamide A-73 [00083]  4-propyl-N-(2- (trifluoromethyl)phenyl) benzenesulfonamide A-74 [00084]  N-(5-chloro-2- methoxyphenyl)-4- propylbenzenesulfonamide A-75 [00085]  3-((4- propylphenyl)sulfonamido) benzoic acid A-76 [00086]  N-(2,6- diisopropylphenyl)-4- propylbenzenesulfonamide A-77 [00087]  N-(3-((4- propylphenyl)sulfonamido) phenyl)acetamide A-78 [00088]  N,N-diethyl-4-((4-

propylphenyl)sulfonamido) benzamide A-79 [00089]  N-cyclohexyl-N-methyl-4-((4-propylphenyl)sulfonamido) benzenesulfonamide A-80 [00090]  N-methyl-4-((4-propylphenyl)sulfonamido) benzenesulfonamide A-81 [00091]  4-((4-propylphenyl)sulfonamido) benzamide A-82 [00092]  N-(2-chloro-4-methylphenyl)-4-propylbenzenesulfonamide A-83 [00093]  N-(4-butylphenyl)-4-propylbenzenesulfonamide A-84 [00094]  3-((4-propylphenyl)sulfonamido) benzamide A-85 [00095]  4-propyl-N-(4-(N-(thiazol-2-yl)sulfamoyl)phenyl)benzene-sulfonamide A-86 [00096]  2-((4-propylphenyl)sulfonamido) benzamide A-87 [00097]  N-(2,6-difluorophenyl)-4-propylbenzenesulfonamide A-88 [00098]  N-(2-methoxy-5-methylphenyl)-4-propylbenzenesulfonamide A-89 [00099]  N,N,2-trimethyl-5-((4-propylphenyl)sulfonamido) benzenesulfonamide A-90 [00100]  N-(2-methyl-5-(piperidin-1-ylsulfonyl)phenyl)-4-propylbenzenesulfonamide A-91 [00101]  N-(2-benzylphenyl)-4-propylbenzenesulfonamide A-92 [00102]  N-(2,6-dimethylphenyl)-4-propylbenzenesulfonamide A-93 [00103]  4-propyl-N-(4-(6,7,8,9-tetrahydro-5H-[1,2,4]triazolo[4,3-a]azepin-3-yl)phenyl) benzenesulfonamide A-94 [00104]  4-fluoro-N-(2-((4-propylphenyl)sulfonamido)phenyl)benzenesulfonamide A-96 [00105]  methyl 4,5-dimethoxy-2-((4-propylphenyl)sulfonamido) benzoate A-97 [00106]  N,N-dimethyl-4-((4-propylphenyl)sulfonamido) benzenesulfonamide A-98 [00107]  4-(3-(4-nitrophenyl)ureido)-N-(m-tolyl)benzenesulfonamide A-99 [00108]  N-(1,2,3,4-tetrahydroquinolin-6-yl)-[1,1'-biphenyl]-4-sulfonamide A-100 [00109]  4-(2-methoxyphenoxy)-N-(m-tolyl)benzenesulfonamide A-101 [00110]  2-(4-(N-(m-tolyl)sulfamoyl)phenoxy) acetic acid A-102 [00111]  2-chloro-5-((4-propylphenyl)sulfonamido) benzoic acid A-103 [00112]  3-(4-(N-(m-tolyl)sulfamoyl)phenyl) propanoic acid A-104 [00113]  N-(4-methyl-3-sulfamoylphenyl)-4-propylbenzenesulfonamide A-106 [00114]  4-([1,1'-biphenyl]-4-ylsulfonyl)-1,4-thiazepane A-107 [00115]  N-(2-(4-methyl-4H-1,2,4-triazol-3-yl)phenyl)-4-propylbenzenesulfonamide A-108 [00116]  3-([1,1'-biphenyl]-4-sulfonamido)-N,N,4-trimethylbenzamide A-109 [00117]  N-(2-(1H-tetrazol-1-yl)phenyl)-4-propylbenzenesulfonamide A-110 [00118]  N-(3-((2-oxooxazolidin-3-yl)methyl)phenyl)-4-propylbenzenesulfonamide A-111 [00119]  N-(4-((1H-pyrazol-1-yl)methyl)phenyl)-4-propylbenzenesulfonamide A-113 [00120]  N-(2-((1H-1,2,4-triazol-1-yl)methyl)phenyl)-4-propylbenzenesulfonamide A-114 [00121]  1-([1,1'-biphenyl]-4-ylsulfonyl)indoline A-115 [00122]  N-(1-acetylindolin-5-yl)-4-propylbenzenesulfonamide A-116 [00123]  N-(3-chloro-2-morpholinophenyl)-[1,1'-biphenyl]-4-sulfonamide A-117 [00124]  N-(1-ethylpiperidin-3-yl)-[1,1'-biphenyl]-4-sulfonamide A-118 [00125]  N-(2-methyl-3-((4-propylphenyl)sulfonamido)phenyl)butyramide A-119 [00126]  N-(2-chloro-5-((4-propylphenyl)sulfonamido)phenyl)acetamide A-120 [00127]  N-(4-(4-methyl-6-oxo-1,6-dihydropyrimidin-2-yl)phenyl)-4-propylbenzenesulfonamide A-122 [00128]  N-(5-chloro-2-(1H-1,2,4-triazol-1-yl)phenyl)-4-propylbenzenesulfonamide A-123 [00129]  N-(2-(thiazol-2-yl)ethyl)-[1,1'-biphenyl]-4-sulfonamide

Target Crops and their Pathogens

[00090] The present disclosure provides formulations and methods for their use in treating crops for pathogens. In one embodiment, a disclosed compound is administered in combination with an agricultural or horticultural pesticide, such as an acaricide, antimicrobial, fungicide, herbicide, insecticide, molluscicide and/or nematocide. Crops that can be treated, include those plagued by various pathogens, including without limitation, bacteria, viruses, fungal pathogens, mites,

nematodes, molluscs, weeds or other pests, as is known to those of ordinary skill in the agricultural arts. By way of example, such agricultural and horticultural crops that can be treated according to the present disclosure include plants, whether genetically modified or not, including their harvested products, such as: cereals; vegetables; root crops; potatoes; trees such as fruit trees, for example banana trees, tea, coffee trees, or cocoa trees; grasses; lawn grass; or cotton.

[0091] Roux and coworkers describe the compounds:

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[0092] as enhancing the ability of certain fungicides to inhibit the growth of different plant-pathogenic fungi. These compounds are referred to herein as “Roux compound 1,” and “Roux compound 13,” respectively (*Molecular Plant Pathology*, 2017, 18(7), 1012-1023; and WO 2016/123191). The present compounds surprisingly enhance the ability of a variety of pesticides against a broad variety of pathogens, including fungal pathogens. In addition, examples of the presently disclosed compound exhibit superior enhancer activity than Roux compounds 1 and 13.

[0093] The agricultural or horticultural enhancer disclosed herein may be applied to each part of plants, such as leaves, stems, patterns, flowers, buds, fruits, seeds, sprouts, roots, tubers, tuberous roots, shoots, or cuttings. The agricultural or horticultural enhancer according to the present disclosure may also be applied to improved varieties/varieties, cultivars, as well as mutants, hybrids and genetically modified embodiments of these plants.

[0094] The agricultural or horticultural treatment described herein may be used to conduct seed treatment, foliage application, soil application, or water application, so as to control various diseases occurring in agricultural or horticultural crops, including flowers, lawns, and pastures. occurring in agricultural or horticultural crops, including flowers, lawns, and pastures.

[0095] The present compounds are useful for potentiating the effects of antimicrobial agents. For example, the present compounds can be used in combination with an antimicrobial agent to combat bacterial and viral infection.

[0096] The present compounds are useful for potentiating the effects of herbicides. For example, the present compounds can be used in combination with one or more herbicide to control weeds or other unwanted vegetation.

[0097] The present compounds are useful for potentiating the effects of insecticides. For example, the present compounds can be used in combination with one or more insecticide to control insect infestation.

[0098] The present compounds are useful for potentiating the effects of acaricides or miticides. For example, the present compounds can be used in combination with one or more acaricidal agent to control mites.

[0099] The present compounds are useful for potentiating the effects of molluscicides. For example, the present compounds can be used in combination with one or more molluscicide to prevent interference of slugs or snails with a crop.

[0100] The present compounds are useful for potentiating the effects of nematocides. For example, the present compounds can be used in combination with one or more nematocide to prevent interference of nematodes with a crop.

[0101] The present compounds are particularly useful for potentiating the effects of fungicides against plant fungal pathogens. Examples of pathogens treated according to the present disclosure include, without limitation, *Botrytis cinerea*, *Colletotrichum graminicola*, *Fusarium oxysporum*, *Sclerotiana sclerotiorum*, *Verticillium dahlia*, *Mycospharella graminicola* and *Sphacelotheca reliana*.

[0102] *Botrytis cinerea* is an airborne plant pathogen with a necrotrophic lifestyle attacking over 200 crop hosts worldwide. It mainly attacks dicotyledonous plant species, including important protein, oil, fiber and horticultural crops, grapes and strawberries and also *Botrytis* also causes secondary soft rot of fruits and vegetables during storage, transit and at the market. Many classes of fungicides have failed to control *Botrytis cinerea* due to its genetic plasticity.

[0103] The genus *Colletotrichum* comprises ~600 species attacking over 3,200 species of monocot and dicot plants. *Colletotrichum graminicola* primarily infects maize (*Zea mays*), causing annual losses of approximately 1 billion dollars in the United States alone (Connell et al., 2012).

[0104] *Fusarium* wilt of banana, caused by the soil-borne fungus *Fusarium oxysporum* f.sp. *cubense*, is a major threat to banana production worldwide. No fungicides are currently available to effectively control the disease once plants are infected (Peng J et al., 2014).

[0105] The white mold fungus *Sclerotinia sclerotiorum* is known to attack more than 400 host species and is considered one of the most prolific plant pathogens. The majority of the affected crop species are dicotyledonous, along with a number of agriculturally significant monocotyledonous plants. Some important crops affected by *S. sclerotiorum* include legumes (soybean), most vegetables, stone fruits and tobacco.

[0106] The ascomycete *Verticillium dahliae* is a soil-borne fungal plant pathogen that causes vascular wilt diseases in a broad range of dicotyledonous host species. *V. dahliae* can cause severe yield and quality losses in cotton and other important crops such as vegetables, fibers, fruit, nut trees, forest trees and ornamental plants.

[0107] The ascomycete fungus *Mycosphaerella graminicola* (anamorph: *Septoria tritici*) is one of the most important foliar diseases of wheat leaves, occurring wherever wheat is grown. Yield losses attributed to this disease range from 25%-50%, and are especially high in Europe, the Mediterranean region and East Africa. Infection by *M. graminicola* is initiated by air borne ascospores produced on residues of last season's crop. Primary infection usually occurs after seedlings emerge in spring or fall. The mature disease is characterized by necrotic lesions on the leaves and stems of infected plants.

[0108] The basidiomycete fungus *Sphacelotheca reiliana* infects corn (*Zea mays*) systemically, causing Head Smut. Yield loss attributed to the disease is variable, and is directly dependent on the incidence of the disease. The fungus overwinters as diploid teliospores in crop debris or soil. Floral structures are converted to sori containing masses of powdery teliospores that resemble mature galls of common smut.

[0109] Examples of crops to be treated and plant diseases (pathogens) to be controlled using the presently disclosed compounds and compositions include, without limitation: Sugar beet: brown spot disease (*Cercospora beticola*), black root disease (*Aphanomyces cochlioides*), root rot disease (*Thanatephorus cucumeris*), leaf rot disease (*Thanatephorus cucumeris*), and the like.

[0110] Peanut: brown spot disease (*Mycosphaerella arachidis*), leaf mold (*Ascochyta* sp.), rust disease (*Puccinia arachidis*), damping-off disease (*Pythium debaryanum*), rust spot disease (*Alternaria alternata*), stem rot disease (*Sclerotium rolfsii*), black rust disease (*Mycosphaerella berkeleyi*), and the like.

[0111] Cucumber: powdery mildew (*Sphaerotheca fuliginea*), downy mildew (*Pseudoperonospora cubensis*), gummy stem blight (*Mycosphaerella melonis*), wilt disease (*Fusarium oxysporum*), sclerotinia rot (*Sclerotinia sclerotiorum*), gray mold (*Botrytis cinerea*), anthracnose (*Colletotrichum orbiculare*), scab (*Cladosporium cucumerinum*), brown spot disease (*Corynespora cassiicola*), damping-off disease (*Pythium debaryanum*, *Rhizoctonia solani* Kuhn), *Phomopsis* root rot disease (*Phomopsis* sp.), Bacterial spot (*Pseudomonas syringae* pv. *Lechrymans*), and the like.

[0112] Tomato: gray mold disease (*Botrytis cinerea*), leaf mold disease (*Cladosporium fulvum*), late blight disease (*Phytophthora infestans*), *Verticillium* wilt disease (*Verticillium albo-atrum*, *Verticillium dahliae*), powdery mildew disease (*Oidium neolycopersici*), early blight disease (*Alternaria solani*), leaf mold disease (*Pseudocercospora fuligena*), and the like.

[0113] Eggplant: gray mold disease (*Botrytis cinerea*), black rot disease (*Corynespora melongenae*), powdery mildew disease (*Erysiphe cichoracearum*), leaf mold disease (*Mycovellosiella natrassii*), sclerotinia rot disease (*Sclerotinia sclerotiorum*), *Verticillium* wilt disease (*Verticillium dahlia*), *Mycosphaerella* blight (*Phomopsis vexans*), and the like.

[0114] Strawberry: gray mold disease (*Botrytis cinerea*), powdery mildew disease (*Sphaerotheca*

humuli), anthracnose disease (*Colletotrichum acutatum*, *Colletotrichum fragariae*), phytophthora rot disease (*Phytophthora cactorum*), soft rot disease (*Rhizopus stolonifer*), fusarium wilt disease (*Fusarium oxysporum*), verticillium disease dahlia), and wilt (*Verticillium the like*).

[0115] Onion: neck rot disease (*Botrytis allii*), gray mold disease (*Botrytis cinerea*), leaf blight disease (*Botrytis squamosa*), downy mildew disease (*Peronospora destructor*), *Phytophthora* porn disease (*Phytophthora porn*), and the like.

[0116] Cabbage: clubroot disease (*Plasmodiophora brassicae*), soft rot disease (*Erwinia carotovora*), black rot disease (*Xanthomonas campestris* pv. *campestris*), bacterial black spot disease (*Pseudomonas syringae* pv. *Maculicola*, P.s. pv. *alisalensis*), downy mildew disease (*Peronospora parasitica*), sclerotinia rot disease (*Sclerotinia sclerotiorum*), black spot disease (*Alternaria brassicicola*), gray mold disease (*Botrytis cinerea*), and the like.

[0117] Common bean: sclerotinia rot disease (*Sclerotinia sclerotiorum*), gray mold disease (*Botrytis cinerea*), anthracnose (*Colletotrichum lindemuthianum*), angular spot disease (*Phaeoisariopsis griseola*), and the like.

[0118] Apple: powdery mildew disease (*Podosphaera leucotricha*), scab disease (*Venturia inaequalis*), Monilinia disease (*Monilinia mali*), black spot disease (*Mycosphaerella pomi*), valla canker disease (*Valsa mali*), *alternaria* blotch disease (*Alternaria mali*), rust disease (*Gymnosporangium yamadae*), ring rot disease (*Botryosphaeria berengeriana*), anthracnose disease (*Glomerella cingulata*, *Colletotrichum acutatum*), leaf rot disease (*Diplocarpon mali*), fly speck disease (*Zygophiala jamaicensis*), Sooty blotch (*Gloeodes pomigena*), violet root rot disease (*Helicobasidium mompa*), gray mold disease (*Botrytis cinerea*), and the like.

[0119] Japanese apricot: scab disease (*Cladosporium carpophilum*), gray mold disease (*Botrytis cinerea*), brown rot disease (*Monilinia mumecola*), and the like.

[0120] Persimmon: powdery mildew disease (*Phyllactinia kakicola*), anthracnose disease (*Gloeosporium kaki*), angular leaf spot (*Cercospora kaki*), and the like.

[0121] Peach: brown rot disease (*Monilinia fructicola*), scab disease (*Cladosporium carpophilum*), *phomopsis* rot disease (*Phomopsis* sp.), bacterial shot hole disease (*Xanthomonas campestris* pv. *pruni*), and the like.

[0122] Almond: brown rot disease (*Monilinia taxa*), spot blotch disease (*Stigmina carpophila*), scab disease (*Cladosporium carpophilum*), red leaf spot disease (*Polystigma rubrum*), *alternaria* blotch disease (*Alternaria alternata*), anthracnose (*Colletotrichum gloeosporoides*), and the like.

[0123] Yellow peach: brown rot disease (*Monilinia fructicola*), anthracnose disease (*Colletotrichum acutatum*), black spot disease (*Alternaria* sp.), *Monilinia kusanoi* disease (*Monilinia kusanoi*), and the like.

[0124] Grape: gray mold disease (*Botrytis cinerea*), powdery mildew disease (*Uncinula necator*), ripe rot disease (*Glomerella cingulata*, *Colletotrichum acutatum*), downy mildew disease (*Plasmopara viticola*), anthracnose disease (*Elsinoe ampelina*), brown spot disease (*Pseudocercospora vitis*), black rot disease (*Guignardia bidwellii*), white rot disease (*Coniella castaneicola*), rust disease (*Phakopsora ampelopsidis*), and the like.

[0125] Pear: scab disease (*Venturia nashicola*), rust disease (*Gymnosporangium asiaticum*), black spot disease (*Alternaria kikuchiana*), ring rot disease (*Botryosphaeria berengeriana*), powdery mildew disease (*Phyllactinia mali*), Cytospora canker disease (*Phomopsis fukushii*), brown spot blotch disease (*Stemphylium vesicarium*), anthracnose disease (*Glomerella cingulata*), and the like.

[0126] Tea: ring spot disease (*Pestalotiopsis longiseta*, *P. theae*), anthracnose disease (*Colletotrichum theae-sinensis*), Net blister blight (*Exobasidium reticulatum*), and the like.

[0127] Citrus fruits: scab disease (*Elsinoe fawcettii*), blue mold disease (*Penicillium italicum*), common green mold disease (*Penicillium digitatum*), gray mold disease (*Botrytis cinerea*), melanose disease (*Diaporthe citri*), canker disease (*Xanthomonas campestris* pv. *Citri*), powdery mildew disease (*Oidium* sp.), and the like.

[0128] Wheat: powdery mildew (*Blumeria graminis* f. sp. *tritici*), red mold disease (*Gibberella*

zeae), brown rust disease (*Puccinia recondita*), brown snow mold disease (*Pythium iwayamai*), pink snow mold disease (*Monographella nivalis*), eye spot disease (*Pseudocercospora herpotrichoides*), leaf scorch disease (*Septoria tritici*), glume blotch disease (*Leptosphaeria nodorum*), typhula snow blight disease (*Typhula incarnata*), sclerotinia snow blight disease (*Myriosclerotinia borealis*), damping-off disease (*Gaeumannomyces graminis*), ergot disease (*Claviceps purpurea*), stinking smut disease (*Tilletia caries*), loose smut disease (*Ustilago nuda*), and the like.

[0129] Barley: leaf spot disease (*Pyrenophora graminea*), net blotch disease (*Pyrenophora teres*), leaf blotch disease (*Rhynchosporium secalis*), loose smut disease (*Ustilago tritici*, *U. nuda*), and the like.

[0130] Rice: blast disease (*Pyricularia oryzae*), sheath blight disease (*Rhizoctonia solani*), bakanae disease (*Gibberella fujikuroi*), brown spot disease (*Cochliobolus miyabeanus*), damping-off disease (*Pythium graminicola*), bacterial leaf blight (*Xanthomonas oryzae*), bacterial seedling blight disease (*Burkholderia plantarii*), brown stripe disease (*Acidovorax avenae*), bacterial grain rot disease (*Burkholderia glumae*), *Cercospora* leaf spot disease (*Cercospora oryzae*), false smut disease (*Ustilaginoidea virens*), rice brown spot disease (*Alternaria alternata*, *Curvularia intermedia*), kernel discoloration of rice (*Alternaria padwickii*), pink coloring of rice grains (*Epicoccum purpurascens*), and the like.

[0131] Tobacco: sclerotinia rot disease (*Sclerotinia sclerotiorum*), powdery mildew disease (*Erysiphe cichoracearum*), phytophthora rot disease (*Phytophthora nicotianae*), and the like.

[0132] Tulip: gray mold disease (*Botrytis cinerea*), and the like.

[0133] Sunflower: downy mildew disease (*Plasmopara halstedii*), sclerotinia rot disease (*Sclerotinia sclerotiorum*), and the like.

[0134] Bent grass: Sclerotinia snow blight (*Sclerotinia borealis*), Large patch (*Rhizoctonia solani*), Brown patch (*Rhizoctonia solani*), Dollar spot (*Sclerotinia homoeocarpa*), blast disease (*Pyricularia* sp.), *Pythium* red blight disease (*Pythium aphanidermatum*), anthracnose disease (*Colletotrichum graminicola*), and the like.

[0135] Orchard grass: powdery mildew disease (*Erysiphe graminis*), and the like.

[0136] Soybean: purple stain disease (*Cercospora kikuchii*), downy mildew disease (*Peronospora manshurica*), phytophthora rot disease (*Phytophthora sojae*), rust disease (*Phakopsora pachyrhizi*), sclerotinia rot disease (*Sclerotinia sclerotiorum*), anthracnose disease (*Colletotrichum truncatum*), gray mold disease (*Botrytis cinerea*), Sphaceloma scab (*Elsinoe glycines*), melanoses (*Diaporthe phaseolorum* var. *sojae*), and the like.

[0137] Potato: hytophthora rot disease (*Phytophthora infestans*), early blight disease (*Alternaria solani*), scurf disease (*Thanatephorus cucumeris*), verticillium wilt disease (*Verticillium albo-atrum*, *V. dahlia*, *V. nigrescens*, and the like.

[0138] Banana: Panama disease (*Fusarium oxysporum*), Sigatoka disease (*Mycosphaerella fijiensis*, *M. musicola*), and the like.

[0139] Rapeseed: sclerotinia rot disease (*Sclerotinia sclerotiorum*), root rot disease (*Phoma lingam*), black leaf spot disease (*Alternaria brassicae*), and the like.

[0140] Coffee: rust disease (*Hemileia vastatrix*), anthracnose (*Colletotrichum coffeanum*), leaf spot disease (*Cercospora coffeicola*), and the like.

[0141] Sugarcane: brown rust disease (*Puccinia melanocephala*), and the like.

[0142] Corn: zonate spot disease (*Gloeocercospora sorghi*), rust disease (*Puccinia sorghi*), southern rust disease (*Puccinia polysora*), smut disease (*Ustilago maydis*), brown spot disease (*Cochliobolus heterostrophus*), northern leaf blight (*Setosphaeria turcica*), and the like.

[0143] Cotton: seedling blight disease (*Pythium* sp.), rust disease (*Phakopsora gossypii*), sour rot disease (*Mycosphaerella areola*), anthracnose (*Glomerella gossypii*), and the like.

Pesticides

[0144] The presently disclosed compounds are useful for enhancing the effect of a variety of

agrochemicals, including fungicides, antiviral agents, bactericides, herbicides, insecticidal/acaricidal agents, molluscicides, nematocides, soil pesticides, plant control agents, synergistic agents, fertilizers and soil conditioners.

[0145] In one embodiment, the presently disclosed compounds are useful for enhancing the fungicidal effect of a variety of fungicides. Fungicides for use with the presently disclosed compounds are well known to those of skill in the art and include, without limitation, those set forth by class in Table 2, which is shown below.

TABLE-US-00002 TABLE 2 Family & Group # Common Names Trade Names (Combination Products)

Benzimidazole (Group 1)	benomyl Benlate, Tersan 1991 thiabendazole Arbotect 20-S, Decco Salt No. 19, LSP Flowable Fungicide, Mertect 340-F thiophanate-methyl Cavalier, Cleary's 3336, OHP 6672, Regal SysTec, Tee-Off, T-Methyl 4.5F AG, TM 85, Topsin M Dicarboximide (Group 2)
iprodione Epic 30, Ipro, Meteor, Nevado, OHP Chipco 26019, Rovral, (Interface)	vinclozolin Curalan, Ronilan Phenylpyrroles (Group 12)
fludioxonil Cannonball, Emblem, Maxim, Medallion, Mozart, Scholar, Spirato, (Academy, Miravis Prime, Palladium, Switch)	Anilinopyrimidines (Group 9)
cyprodinil Vanguard (Palladium, Switch, Inspire Super) pyrimethanil Penbotec, Scala, (Luna Tranquility) Hydroxylanilide (Group 17)	fenhexamid Decree, Elevate, Judge fenpyrazamine Protexio Carboxamide (Group 7)
boscalid Emerald, Endura, (Encartis, Honor, Pageant, Pristine) carboxin Vitavax fluopyram Luna Privilege, Velum Prime (Broadform, Luna Experience, Luna Sensation, Luna Tranquility, Propulse) flutolanil Contrast, Moncut, ProStar fluxapyroxad (Lexicon, Merivon, Orkestra) inpyrfluxam Excalia isofetamid Kenja oxycarboxin Carboject, Plantvax penthiopyrad Fontelis, Velist, Vertisan pydiflumetofen Miravis, Posterity, Miravis Ace A (Miravis Neo, Miravis Prime, Miravis Duo, Miravis Top) solatenol Aprovia (Contend A, Elatus, Mural) (benzovindiflupyr) Phenylamide (Group 4)	
mefenoxam Apron, Ridomil Gold, Subdue MAXX, (Quadris Ridomil Gold, Uniform) metalaxyl Acquire, Allegiance, MetaStar, Ridomil, Sebring, Subdue oxadixy1 Anchor Phosphonate (Group P7)	aluminum tris Aliette, Flanker, Legion, Signature, Areca Phosphorous Acid Agri-Fos, Alude, Appear, Fiata, Fosphite, Phospho Jet, Phostrol, Rampart, Reload Cinnamic acid (Group 40)
dimethomorph Forum, Stature, (Orvego, Zampro) mandipropamid Micora, Revus, (Revus Top) OSBPI (Group 49)	oxathiapiprolin Segovis Triazoles carboxamide (Group 22)
ethaboxam V-10208 Group 27 cymoxanil Curzate, (Tanos) Carbamate (Group 28)	propamocarb Banol, Previcur, Proplant, Tattoo Benzamide (Group 43)
fluopicolide Adorn, Presidio Demethylation-inhibiting (Group 3)	Piperazines triforine Funginex, Triforine Pyrimidines fenarimol Focus, Rubigan, Vintage Imidazole imazalil Fungaflor, (Raxil MD Extra) triflumizole Procure, Terraguard, Trionic Triazoles cyproconazole Sentinel difenoconazole Dividend, Inspire, (Academy, Briskway, Contend A, Inspire Super, Quadris Top, Revus Top) Miravis Duo fenbuconazole Enable, Indar flutriafol Topguard, (Topguard EQ) mefentrifluconazole Maxtima (Navicon) metconazole Quash, Tourney ipconazole Rancona myclobutanil Eagle, Hoist, Immunox, Laredo, Nova, Rally, Sonoma, Systhane propiconazole Alamo, Banner, Break, Bumper, Infuse, Kestrel Mex, Miravis Ace B, PropiMax, ProPensity, Strider, Tilt, Topaz, (Aframe Plus, Concert, Contend B, Headway, Quilt Xcel, Stratego) prothioconazole Proline (Propulse) tebuconazole Bayer Advanced, Elite, Folicur, Lynx, Mirage, Orius, Raxil, Sativa, Tebucon, Tebuject, Tebusha, Tebustar, Toledo, (Absolute, Luna Experience, Unicorn), etc. tetraconazole Mettle triadimefon Bayleton, Strike, (Armada, Tartan, Trlgo) triadimenol Baytan triticonazole Charter, Trinity, (Pillar) Morpholine (Group 5)
piperalin Pipron spiroxamine Accrue Group U6 cyflufenamid Torino Group 50 metrafenone Vivando pyriofenone Prolivo QoI Strobilurins (Group 11)	azoxystrobin Abound, Aframe, Dynasty, Heritage, Protété, Quadris, Quilt, (Aframe Plus, Briskway, Contend B, Dexter Max, Elatus, Headway, Mural, Quadris Top, Quilt Xcel, Renown, Topguard EQ, Uniform) femoxadone (Tanos) fenamidone Fenstop, Reason fluoxastrobin Aftershock, Disarm, Evito, Fame kresoxim-methyl Cygnus, Sovran mandestrobin Intuity, Pinpoint picoxystrobin Aproach pyraclostrobin Cabrio, Empress, Headline, Insignia, Stamina, (Honor, Lexicon, Merivon, Navicon, Orkestra, Pageant, Pillar, Pristine)

trifloxystrobin Compass, Flint, Gem, (Absolute, Armada, Broadform, Interface, Luna Sensation, Stratego, Tartan, Trigo) Quinoline (Group 13) quinoxifen Quintec Inorganic Compounds Coppers (Group MI) bordeaux None copper ammonium complex Copper Count-N copper hydroxide Champ, Champion, Kalmor, Kentan, Kocide, Nu-Cop copper oxide Nordox copper oxychloride C—O—C—S, Oxycop copper sulfate Cuprofix Disperss, many others Sulfur (Group M2) sulfur Cosavet, Kumulus, Microthiol Disperss, Thiosperse Lime sulfur Ca polysulfides Lime Sulfur, Sulforix Ethylenebisdithiocarbamates mancozeb Dithane, Fore, Penncozeb, Protect, Manex, Manzate, (EBDC) (Group M3) Roper, Wingman, (Dexter Max, Gavel) maneb Maneb metiram Polyram EBDC-like (Group M3) ferbam Carbamate, Ferbam thiram Difiant, Spotrete, Thiram ziram Ziram Aromatic Hydrocarbon (Group 14) dicloran (DCNA) Allisan, Botran etridizole Terrazole, Truban pentachloronitrobenzene Autilus, Defend, Engage, PCNB, Terraclor, (Premion) Chloronitrile (Group M5) chlorothalonil Bravo, Daconil, Docket, Echo, Ensign, Exotherm Termil, Funginil, Legend, Manicure, Pegasus, Terranil, (Concert, Spectro) Phthalimides (Group M4) captan Captan Guanidines (Group U12) dodine Syllit QiI fungicides (Group 21) cyazofamid Ranman, Segway Polyoxin (Group 19) polyoxin Affirm, Endorse, Oso, Ph-D, Tavano, Veranda Group 29 fluazinam Omega, Secure Thiazolidine (U13) flutianil Gatten

[0146] Fungicides are cataloged more broadly by the Fungicide Resistance Action Committee (FRAC) in the FRAC Code List 2022 and reproduced in Appendix 1 and which is incorporated herein by reference in its entirety.

[0147] In one embodiment, a presently disclosed enhancer compound is used in combination with one or more compound from the Families or Groups set forth in Table 2, Appendix 1, or both. In certain embodiments, a presently disclosed enhancer is used in combination with one or more fungicides recited in column 1 of Table 2.

[0148] In particular embodiments, a disclosed enhancer is used in combination with one or more of a fungicide selected from the benzimidazoles, dicarboximides, phenylpyrroles, anilinopyrimidines, hydroxyanilides, carboxamides, phenyl amides, phosphonates, cinnamic acids, oxysterol binding protein inhibitors (OSBPI), triazole carboxamides, cymoxanil, carbamates, benzamides, demethylation inhibiting piperazines, demethylation inhibiting pyrimidines, demethylation inhibiting azoles, including imidazoles, and triazoles, such as cyproconazole, difenoconazole, fenbuconazole, flutriafol, mefentrifluconazole, metconazole, ipconazole, prothioconazole, tebuconazole, tetraconazole, triadimefon, triadimenol, triticonazole, morpholines, cyflufenamid, metrafenone, pyriofenone, strobilurins, copper ammonium complex, copper hydroxide, copper oxide, copper oxychloride, copper sulfate, sulfur, lime sulfur, ethylenebisdithiocarbamates, aromatic hydrocarbons, phthalimides, guanidines, polyoxins, fluazinam and thiazolidines.

[0149] Particular fungicides that are potentiated by use in combination with an enhancer according to the methods herein by administration of an apyrase inhibitor are coppers, such as copper octanoate, copper hydroxide and the like, myclobutanil, propiconazole, tebuconazole, epoxiconazole, difenoconazole, triticonazole, and prothioconazole.

[0150] In one embodiment, the combined treatment with a selected fungicide and an enhancer according to the present disclosure provides synergistic fungicidal activity against plant pathogenic fungi.

[0151] In one embodiment, the disclosure provides compositions and methods of treating plants or plant seeds infected with or at risk of being infected with a fungal pathogen. In one embodiment compositions of the present disclosure comprise a formulation of a fungicide, an enhancer and a phytologically acceptable carrier. In another embodiment, the fungicide and enhancer are administered in separate compositions. In further embodiments, an agricultural or horticultural fungicide is used in combination with other compounds in addition to the presently disclosed apyrase inhibitors. As with the apyrase inhibitors, such other compounds can be administered in the same or separate compositions as the fungicide. Examples of the other components include known carriers to be used to conduct formulation. Additional examples thereof include conventionally-

known herbicides, insecticidal/acaricidal agents, nematodes, soil pesticides, plant control agents, synergistic agents, fertilizers, soil conditioners, and animal feeds. In one embodiment, the inclusion of such other components yields synergistic effects on crop growth.

[0152] In one embodiment, the presently disclosed compounds are used to potentiate the effect of a herbicide. Exemplary herbicides for use in combination with the present compounds are known to those of skill in the art and include, without limitation, those described in Appendix 2. By way of example, suitable herbicides for use in combination with the present compounds include inhibitors of acetyl COA synthase, inhibitors of acetolactate synthesis, inhibitors of microtubule assembly, inhibitors of microtubule organization, auxin mimics, photosynthesis inhibitors, deoxy-D-xylulose phosphate synthase inhibitors, enolpyruvyl shikimate phosphate synthase inhibitors, phytoene desaturase inhibitors, glutamine synthetase inhibitors, dihydropteroate synthesis inhibitors, protoporphyrinogen oxidase inhibitors, cellulose synthesis inhibitors, uncouplers, hydroxyphenyl pyruvate dioxygenase inhibitors, fatty acid thioesterase inhibitors, serine-threonine protein phosphatase inhibitors, solanesyl diphosphate synthase inhibitors, inhibitors of very long-chain fatty acid synthesis, homogentisate solanesyltransferase inhibitors, lycopene cyclase inhibitors.

[0153] In one embodiment, the presently disclosed compounds are used to potentiate the effect of an insecticide. Exemplary insecticides for use in combination with the present compounds are known to those of skill in the art and include, without limitation, those described in Appendix 3.

FORMULATIONS

Formulations

[0154] The present disclosure provides specific apyrase inhibitors to enhance the potency of pesticides to effectively restrict the growth of plant pathogenic species. In certain non-limiting embodiments, the apyrase inhibitors can be provided at: from about 0.01 to about 80% weight to weight in a final composition, or from about 25% to about 55%, such as from about 30% to about 50%, from about 35% to about 45%, such as about 0.01, 0.05, 0.1, 0.5, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.5, 3.0, 4.0, 5.0, 7.5, 10, 20, 30, 40, 50, 55, 60 or 80% weight to weight in a final composition. In one embodiment the apyrase inhibitors are provided in liquid form at from about 0.01 to about 50%, such as from about 15% to about 50%, from about 20% to about 45%, from about 25% to about 40%, such as about 0.01, 0.05, 0.1, 0.5, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.5, 3.0, 4.0, 5.0, 7.5, 10, 15, 20, 30, 40 or 50% volume to volume in a final diluted composition. The skilled artisan will recognize that the formulation of the pesticide, the apyrase inhibitor or a combination thereof can be provided in a concentrate that can be diluted prior to use, or can be provided in a diluted form ready for treatment.

[0155] The enhancer, pesticide and combinations thereof are not particularly limited by the dosage form. Examples of the dosage form include wettable powders, emulsions, emulsifiable concentrates, oil-dispersible liquids, powders, granules, water-soluble agents, suspensions, granular wettable powders, and tablets. The method for preparing formulation is not particularly limited, and conventionally-known methods may be adopted depending on the dosage form.

[0156] Several formulation examples are described below. The preparation formulations shown below are merely examples, and may be modified within a range not contrary to the essence of the present disclosure. For example, additional active and inert components may be added to the formulations below.

[0157] “Part” means “part by mass” unless otherwise specified.

Formulation Example 1: Wettable Powders

[0158] 40 parts of an enhancer disclosed herein, 53 parts of diatomaceous earth, 4 parts of ethoxylated higher alcohol sulfate ester combined with a suitable solid carrier such as magnesium sulfate, and 3 parts of alkyl naphthalene sulfonate are mixed uniformly, and then finely pulverized to obtain wettable powders containing 40 parts by mass of the enhancer.

Formulation Example 2: Emulsifiable Concentrates

[0159] 3 parts of an enhancer disclosed herein, 60 parts of mixed petroleum distillates, 27 parts of

dimethyl lactamide, and 10 parts of tristyrilphenol ethoxylates are mixed and dissolved to obtain an emulsifiable concentrate containing 3% by mass of the enhancer.

Formulation Example 3: Granules

[0160] 5 parts of an enhancer disclosed herein, 10 parts of talc, 38 parts of clay, 10 parts of bentonite, 30 parts of sodium lignosulfonate and 7 parts of sodium alkyl sulfate are mixed uniformly, and then finely pulverized, followed by conducting fluidized bed granulation to make the median particle diameter thereof be 0.2 to 2.0 mm, and thus granules containing 5% by mass of an enhancer on a dry weight basis disclosed herein are obtained.

Formulation Example 4: Granules

[0161] 5 parts of an enhancer disclosed herein, 73 parts of clay, 20 parts of bentonite, 1 part of sodium dioctyl sulfosuccinate, and 1 part of potassium phosphate are mixed and then pulverized, followed by adding water thereto, and then kneading the mixture. Then, extrusion granulation is conducted, and the resultant is dried to obtain granules containing 5% by mass of the enhancer on a dry weight basis.

Formulation Example 5: Suspensions

[0162] 10 parts of an enhancer disclosed herein, 4 parts of polyoxyethylene alkyl ether, 2 parts of 3 kDa sodium polycarboxylate as dispersant, 10 parts of glycerin, 0.2 parts of xanthan gum, 0.1 parts of biocides as stabilizer, 0.1 parts of organosilicone antifoam emulsion and 73.6 parts of water are mixed, and then wet pulverized until the particle size is 3 microns or less to obtain a suspension containing 10% by mass of the enhancer.

Formulation Example 6: Oil Dispersible Concentrates

[0163] 40 parts of an enhancer disclosed herein, 5 parts of Atlox 4914, 5 parts of organo-modified bentonite and 50 parts of methylated rapeseed oil as carrier are mixed uniformly and then wet pulverized until the median particle size is 3 microns or less to obtain an oil dispersible concentrate containing 40% by mass of the enhancer.

[0164] The skilled artisan will recognize that the various compositions are used commercially at varying concentrations and formulations. For example, it is common for fungicides to be formulated as liquids commercially at 10-40% concentrations. In one embodiment, the presently disclosed enhancers allow the use of a lower amount of a given fungicide due to the enhanced efficacy of fungicide in combination with an enhancer disclosed herein.

Methods for Assessing Enhanced Activity

Apyrase Inhibition Assay

[0165] Apyrase inhibitors useful as enhancers of pesticidal activity are assessed using an in vitro assay. The method of Windsor, *Bio Techniques* 33:1024-1030 (November 2002) was used as follows

Screen for Apyrase Inhibitors—

[0166] 96 well plates were used for the assay: (Greiner bio-one: REF-655901-96 well, PS, F-bottom, Clear, Non-binding)

Buffers:

[0167] Reaction Buffer: 60 mM Hepes; 3 mM MgCl.sub.2, 3 mM CaCl.sub.2) and 3 mM ATP (pH 6.5) [0168] Development Buffer A: 2% aqueous ammonium molybdate [0169] Development Buffer B: 11% ascorbic acid in 37.5% TCA in water [0170] Stop buffer C: 2% trisodium citrate in 2% acetic acid solution in water

Steps:

[0171] Add 100 µl of reaction buffer to each well. [0172] Add 10 µl of DMSO (control) or inhibitor/compound or compounds such as N1915 or orthovanadate to each well. (use inhibitor conc at 1 mM; orthovanadate at 2 mM and N1915 at 1 mM) [0173] Add 10 µl of apyrase (concentration based on optimization—Dilute 1 U/µl enzyme to different concentrations such as 0.1 U, 0.05 U, 0.0025, 0.001 U, 0.0005 U—to find a good range) [0174] Incubate plate at room temperature for 1 hr [0175] Mix development buffer A and B in the ratio of 1:1.5 (just before use).

[0176] Add 50 µl of A:B mix in each well (incubate for 2 mins) [0177] Add 50 µl of C in each well [0178] Measure/Read Absorbance of plate @630 nm [0179] Inhibitory data for the apyrase assay described above are provided for selected compounds in Table 3. D is inhibition of 0% to 10%, C is inhibition from 10% to 20%, B is inhibition from 20% to 30%, and A is inhibition above 30%. Blank entries signify compounds that were not tested.

TABLE-US-00003

TABLE 3 Compound Inhibition	
A-1	C
A-3	A
A-4	B
A-5	A
A-6	D
A-7	C
A-8	A
A-9	A
A-10	A
A-11	D
A-12	C
A-13	D
A-14	D
A-15	D
A-16	C
A-18	C
A-19	B
A-20	D
A-21	A
A-22	D
A-23	B
A-24	C
A-25	D
A-26	B
A-27	D
A-28	D
A-29	A
A-30	C
A-31	D
A-32	C
A-33	B
A-34	A
A-35	D
A-36	A
A-37	B
A-38	C
A-39	D
A-40	D
A-41	A
A-42	B
A-43	C
A-44	D
A-45	C
A-46	A
A-47	B
A-48	B
A-49	D
A-50	C
A-51	D
A-52	C
A-53	C
A-54	D
A-55	D
A-56	C
A-57	D
A-58	D
A-59	A
A-60	D
A-61	D
A-62	A
A-63	C
A-64	A
A-65	B
A-66	C
A-67	C
A-68	D
A-69	D
A-70	C
A-71	D
A-72	C
A-73	D
A-74	C
A-75	D
A-76	C
A-77	D
A-78	D
A-79	A
A-80	D
A-81	D
A-82	C
A-83	B
A-84	D
A-85	D
A-86	D
A-87	C
A-88	D
A-89	D
A-90	A
A-91	B
A-92	B
A-93	D
A-94	A
A-96	C
A-97	C
A-98	A
A-99	A
A-100	D
A-101	B
A-102	D
A-103	B
A-104	D
A-106	D
A-107	C
A-108	A
A-109	C
A-110	C
A-111	D
A-113	D
A-114	D
A-115	A
A-116	A
A-117	D
A-118	D
A-119	D
A-120	B
A-122	C
A-123	D

Method 2: In Vitro Assessment of Combination Activity

[0180] Selected compounds are assessed in combination with fungicides against a range of commercially important plant pathogenic fungi.

[0181] The test is conducted as follows. A fungicide is applied to a fungal plant pathogen at a rate slightly below that at which it gave any control, in combination with a suitable dose of the test compound. The test compounds are recorded as active if control of the pathogen was observed.

[0182] In more detail, the test is conducted as follows. For each combination of fungicide, pathogen and test compound, the following wells are used. Well 1 contains a fungal pathogen growing on agar, and a fungicide at a rate just below that at which it gave any control of the pathogen. Well 2 is the same as Well 1, except that the test compound is also added at Rate 1. Well 3 is the same as Well 2, except that the test compound is added at Rate 2, where Rate 2 is higher than Rate 1. Finally, as a benchmark, Well 4 is the same as Well 1, except that it contained the fungicide at a higher rate, at which it gave partial control of the pathogen. Each of the Wells 1 to 4 are run in duplicate, giving a total of 8 wells for each combination of fungicide, pathogen and test compound. For each well, after a suitable period of incubation, a visual assessment of the % control of the pathogen by the fungicide is made. Test compounds are scored as inactive, active or highly active.

[0183] The following fungicides are used in this assay: azoxystrobin, fluxapyroxad, and desthio prothioconazole. The following fungal pathogens are used in this assay: First, a strain of *Zymoseptoria tritici* with a reduced susceptibility to strobilurin fungicides; second a strain of *Zymoseptoria tritici* with a reduced susceptibility to SDHI fungicides (i.e., those that inhibit succinate dehydrogenase); and third, *Microdochium nivale*. In this assay, Roux Compound 13 exhibited activity in only one combination, enhancing the activity of azoxystrobin against *Microdochium nivale*, but failing to enhance activity of any fungicide against either of the *Zymoseptoria tritici* strains. In contrast, compounds of the present disclosure are effective in the combination assay, enhancing fungicidal activity of one or more fungicides against at least one strain.

Method 3: Greenhouse Crop Tests

[0184] In this method, exemplary compounds were evaluated for their ability to control Brown Rust (*Puccinia recondita*) on wheat in a controlled greenhouse environment in combination with one of four fungicides, Amistar, Imtrex, Proline or Balaya. In these studies, wheat plants (JB Diego) were used. Seeds were sown in 9 cm diameter pots to a depth of 1 to 2 cm using Petersfield potting compost (75% medium grade peat, 12% screened sterilized loam, 3% medium grade vermiculite, 10% grit (5 mm screened, lime free), 1.5 kg PG mix per m³, lime to pH5.5-6.0 and

forms and variations. In some cases, the aryl ring can be substituted with an R group at the ortho or meta position, or substituted with multiple R groups at multiple positions.

##STR00131## [0192] wherein: [0193] n and m are each independently 0, 1, 2, or 3, provided that n plus m is less than or equal to 3.

[0194] R.sup.a and R.sup.b are as described herein. In some cases R.sup.a is phenyl, alkyl, heteroaryl, or cycloalkyl.

[0195] Such syntheses are reported in Journal of Medicinal Chemistry 2022, 65, 15710-15724; Bioorganic & Medicinal Chemistry Letters 2020, 30, 127650; and Journal of Medicinal Chemistry 2004, 47, 4979-4982.

[0196] Many sulfonyl chloride ($\text{—SO}_2\text{Cl}$) starting compounds that can be used in the above scheme are commercially available, or can be chemically synthesized from commercially available compounds. For example, the compound 4-biphenylsulfonyl chloride (CAS number 1623-93-4) is an exemplary starting material where m is 0, n is 1, and the one R.sup.a group is phenyl. 4-biphenylsulfonyl chloride is commercially sold by more than 60 chemical suppliers.

Appendix 1

[0197] Shown below is Appendix 1

TABLE-US-00006 CHEMICAL OR TARGET SITE BIOLOGICAL FRAC MOA AND CODE
GROUP NAME GROUP COMMON NAME COMMENTS CODE A:

text missing or illegible when filed A1 PA - fungicides acylalanines benalaxyl Resistance and cross 4 RNA polymerase I (PhenylAmides) benalaxyl-M resistance well known (=kiralaxyl) in various Oomycetes furalaxyl but mechanism unknown. metalaxyl High risk. metalaxyl-M See FRAC Phenylamide (=mefenoxam) Guidelines for oxazolidinones oxadixyl resistance management butyrolactones ofurace A2 hydroxy- hydroxy- bupirimate Medium risk. Resistance 8 adenosin- (2-amino-) (2-amino-) dimethirimol and cross resistance known deaminase pyrimidines pyrimidines ethirimol in powdery mildews. Resistance management required. A3 heteroaromatics isoxazoles hymexazole Resistance not known. 32 DNA/RNA synthesis isothiazolones othilinone (proposed) A4 carboxylic acids carboxylic acids oxolinic acid Bactericide. Resistance 31 DNA topoisomerase known. type II (gyrase) Risk in fungi unknown. Resistance management required. A5 DHODHI-phenyl-propanol ipflufenquin Medium to high risk. 52 inhibition of fungicides dihydroorotate dehydrogenase within de novo pyrimidine biosynthesis B: B1 MBC - benzimidazoles benomyl Resistance common in many 1 Cytoskeleton tubulin fungicides carbendazim fungal species. Several and motor polymerization (Methyl fuberidazole target site mutations, protein Benzimidazole thiabendazole mostly E198A/G/K, F200Y Carbamates) thiophanates thiophanate in β -tubulin gene. thiophanate-methy Positive cross resistance between the group members. Negative cross resistance to N-phenyl carbamates. High risk. See FRAC Benzimidazole Guidelines for resistance management. B2 N-phenyl N-phenyl diethofencarb Resistance known. Target 10 tubulin carbamates carbamates site mutation E198K. polymerization Negative cross resistance to benzimidazoles. High risk. Resistance management required. B3 benzamides toluamides zoxamide Low to medium risk. 22 tubulin thiazole ethylamino-thiazole- ethaboxam Resistance management polymerization carboxamide carboxamide required. B4 phenylureas phenylureas pencycuron Resistance not known. 20 cell division (unknown site) B5 benzamides pyridinylmethyl-fluopicolide Resistant isolates 43 delocalisation of benzamides fluopimomide detected in grapevine spectrin-like downy mildew. proteins Medium risk. Resistance management required B6 cyanoacrylates aminocyanoacrylates phenamacril Resistance known in 47 actin/myosin/fimbrin *Fusarium graminearum*. function Target site mutations in the gene coding for myosin- 5 found in lab studies. Medium to high risk. Resistance management required. aryl-phenyl- benzophenone metrafenone Less sensitive isolates 50 ketones benzoylpyridine pyriofenone detected in powdery mildews (*Blumeria* and *Sphaerotheca*) Medium risk. Resistance management required. Reclassified from U8 in 2018 B7 pyridazine pyridazine pyridachlometyl High risk. 53 tubulin dynamics modulator C. C1 pyrimidinamines pyrimidinamines diflumetorim Resistance not known. 39

respiration complex I NADH pyrazole-MET1 pyrazole-5- tolpenpyrad oxido-reductase
 carboxamides Quinazoline quinazoline fenazaquin C2 SDHI phenyl-benzamides benodanil
 Resistance known for several 7 complex II: (Succinate- flutolanil fungal species in field succinate-
 dehydro- dehydrogenase mepronil populations and lab mutants. genase inhibitors) phenyl-oxo-
 ethyl isofetamid Target site mutations in *sdh* thiophene amide gene, e.g. H/Y (or H/L) at pyridinyl-
 ethyl- fluopyram 257, 267, 272 or P225L, benzamides dependent on fungal species. phenyl-
 cyclobutyl- cyclobutrifluram Resistance management pyridineamide required. furan- carboxamides
 fenfuram Medium to high risk. oxathiin- carboxin See FRAC SDHI Guidelines carboxamides
 oxycarboxin for resistance management. thiazole- thifluzamide carboxamides pyrazole-4-
 benzovindiflupyr carboxamides bixafen fluindapyr fluxapyroxad furametpyr inpyrfluxam
 isopyrazam penflufen penthiopyrad sedaxane N-cyclopropyl-N- isoflucypram benzyl-pyrazole-
 carboxamides N-methoxy-(phenyl- pydiflumetofen ethyl)-pyrazole- carboxamides pyridine-
 boscaldid carboxamides pyrazine- pyraziflumid carboxamides C. C3 Qol-fungicides methoxy-
 acrylates azoxystrobin Resistance known in various 11 respiration complex III: (Quinone outside
 coumoxystrobin fungal species. Target site cytochrome bc1 Inhibitors) enoxastrobin mutations in
cyt b gene (ubiquinol oxidase) flufenoxystrobin (G143A, F129L) and additional at Qo site (*cyt b*
 picoxystrobin mechanisms. gene) pyraoxystrobin Cross resistance shown methoxy-acetamide
 mandestrobin between all members of the methoxy-carbamates pyraclostrobin Code 11 fungicides.
 pyrametostrobin High risk. triclopyricarb See FRAC Qol Guidelines oximino-acetates kresoxim-
 methyl for resistance management. trifloxystrobin oximino-acetamides dimoxystrobin
 fenaminstrobin metominostrobin orysastrobin oxazolidine-diones famoxadone dihydro-dioxazines
 fluoxastrobin imidazolinones fenamidone benzyl-carbamates pyribencarb Qol-fungicides
 tetrazolinones metyltetraprole Resistance not known. Not 11A (Quinone outside cross resistant
 with Code 11 Inhibitors; fungicides on G143A mutants. Subgroup A) High risk. See FRAC Qol
 Guidelines for resistance management. C: C4 Qil - fungicides cyano-imidazole cyazofamid
 Resistance risk unknown but 21 respiration complex III: (Quinone inside sulfamoyl-triazole
 amisulbrom assumed to be medium to high (continued) cytochrome bc1 Inhibitors) picolinamides
 fenpicoxamid (mutations at target site (ubiquinone florylpicoxamid known in model organisms).
 reductase) at Qi Resistance management site required. No spectrum overlap with the Oomycete-
 fungicides cyazofamid and amisulbrom C5 dinitrophenyl- binapacryl Resistance not known. 29
 uncouplers of crotonates meptyldinocap Also acaricidal activity. oxidative phos- dinocap
 phorylation 2,6-dinitro-anilines fluazinam Low risk. However, resistance claimed in *Botrytis* in
 Japan. (pyr.-hydrazones) (ferimzone) Reclassified to U 14 in 2012. C6 organo tin tri-phenyl tin
 fentin acetate Some resistance cases 30 inhibitors of compounds compounds fentin chloride known.
 Low to medium risk. oxidative phos- fentin hydroxide phorylation, ATP synthase C7 thiophene-
 thiophene- silthiofam Resistance reported. 38 ATP transport carboxamides carboxamides Risk low.
 (proposed) C8 QoSI fungicides triazolo- ametoctradin Not cross resistant to Qol 45 complex III:
 (Quinone outside pyrimidylamine fungicides. cytochrome bc1 Inhibitor, Resistance risk assumed to
 (ubiquinone stigmatellin be medium to high reductase) at binding type) (single site inhibitor). Qo
 site, stigmatellin Resistance management binding sub-site required. D: D1 AP - fungicides anilino-
 pyrimidines cyprodinil Resistance known in *Botrytis* 9 amino methionine (Anilino- mepanipyrim
 and *Venturia*, sporadically acids biosynthesis Pyrimidines) pyrimethanil in *Oculimacula*. and
 (proposed) Medium risk. protein (cgs gene) See FRAC Anilinopyrimidine
text missing or illegible when filed Guidelines for resistance management. D2 enopyranuronic
 enopyranuronic acid blasticidin-S Low to medium risk. 23 protein synthesis acid antibiotic
 antibiotic Resistance management (ribosome, required. termination step) D3 hexopyranosyl
 hexopyranosyl kasugamycin Resistance known in fungal 24 protein synthesis antibiotic antibiotic
 and bacterial (*P. glumae*) (ribosome, initiation pathogens. Medium risk. step) Resistance
 management required. D4 glucopyranosyl glucopyranosyl streptomycin Bactericide. Resistance 25
 protein synthesis antibiotic antibiotic known. High risk. (ribosome, initiation Resistance

management step) required. D5 tetracycline tetracycline oxytetracycline Bactericide. Resistance 41 protein synthesis antibiotic antibiotic known. High risk. (ribosome, Resistance management elongation step) required. E: E1 aza- aryloxyquinoline quinoxifen Resistance to quinoxifen 13 signal signal transduction naphthalenes quinazolinone proquinazid known. transduction (mechanism Medium risk. unknown) Resistance management required. Cross resistance found in *Erysiphe (Uncinula) necator* but not in *Blumeria graminis*. E2 PP-fungicides phenylpyrroles fenpiclonil Resistance found sporadically, 12 MAP/Histidine- (PhenylPyrroles) fludioxonil mechanism speculative. Kinase in osmotic Low to medium risk. signal transduction Resistance management (os-2, HOG1) required. E3 dicarboximides dicarboximides chlozolinate Resistance common in *Botrytis* 2 MAP/Histidine- dimethachlone and some other pathogens. Kinase in osmotic iprodione Several mutations in OS-1, signal transduction procymidone mostly I365S. (os-1, Daf1) vinclozolin Cross resistance common between the group members. Medium to high risk. See FRAC Dicarboximide Guidelines for resistance management F: F1 formerly dicarboximides lipid F2 phosphoro- phosphoro-thiolates edifenphos Resistance known in specific 6 synthesis phospholipid thiolates iprobenfos (IBP) fungi. or biosynthesis, pyrazophos Low to medium risk. transport/ methyltransferase Dithiolanes dithiolanes isoprothiolane Resistance management membrane required if used for risky integrity pathogens. or F3 AH-fungicides aromatic hydrocarbons biphenyl Resistance known in some 14 function cell peroxidation (Aromatic chloroneb fungi. (proposed) Hydrocarbons) dicloran Low to medium risk. (chlorophenyls, quintozone (PCNB) Cross resistance patterns nitroanilines) tecnazene (TCNB) complex due to different tolclofos-methyl activity spectra. heteroaromatics 1,2,4-thiadiazoles etridiazole F4 Carbamates carbamates iodocarb Low to medium risk. 28 cell membrane propamocarb Resistance management permeability, fatty prothiocarb required. acids (proposed) F5 formerly CAA-fungicides F6 formerly *Bacillus amyloliquefaciens* strains (FRAC Code 44); microbial disrupters reclassified to BM02 in 2020 of pathogen cell membranes F7 formerly extract from *Melaleuca alternifolia* (tea tree oil) cell membrane and plant oils (eugenol, geraniol, thymol) disruption FRAC Code 46, reclassified to BM01 in 2021 F8 Polyene amphoteric macrolide natamycin Resistance not known. 48 ergosterol binding antifungal antibiotic (pimaricin) Agricultural, food and topical from *Streptomyces* medical uses. *natalensis* or *S. chattanoogensis* F9 OSBPI piperidinyl-thiazole-oxathiapiprolin Resistance risk assumed to be 49 lipid homeostasis oxysterol binding isoxazolines fluoxapiprolin medium to high (single site and transfer/storage protein inhibitor). Resistance homologue management required. inhibition (Previously U15). F10 protein fragment polypeptide polypeptide Resistance not known. 51 interaction with lipid ASBIOF01-02 fraction of the cell membrane, with multiple effects on cell membrane integrity G: G1 DMI-fungicides piperazines triforine There are big differences in 3 sterol C14- demethylase (DeMethylation pyridines pyrifenoxy the activity spectra of DMI biosynthesis in sterol Inhibitors) pyrisoxazole fungicides. in text missing or illegible when filed biosynthesis (SBI: Class I) pyrimidines fenarimol Resistance is known in various (erg11/cyp51) nuarimol fungal species. Several imidazoles imazalil resistance mechanisms are oxpoconazole known incl. target site pefurazoate mutations in cyp51 (erg 11) prochloraz gene, e.g. V136A, Y137F, triflumizole A379G, I381V; cyp51 triazoles azaconazole promotor; ABC transporters triazolinthiones bitertanol and others. bromuconazole Generally wise to accept that cyproconazole cross resistance is present difenoconazole between DMI fungicides active diniconazole against the same fungus. epoxiconazole DMI fungicides are Sterol etaconazole Biosynthesis Inhibitors (SBIs), fenbuconazole but show no cross resistance fluquinconazole to other SBI classes. flusilazole Medium risk. flutriafol See FRAC SBI Guidelines hexaconazole for resistance management. imibenconazole ipconazole mefentrifluconazole metconazole myclobutanil penconazole propiconazole simeconazole tebuconazole tetraconazole triadimefon triadimenol triticonazole prothioconazole G2 amines morpholines aldimorph Decreased sensitivity for 5 Δ .sup.14-reductase (“morpholines”) dodemorph powdery mildews. and (SBI: Class II) fenpropimorph Cross resistance within the Δ .sup.8.fwdarw. Δ .sup.7- tridemorph group generally

found but not isomerase piperidines fenpropidin to other in sterol piperalin SBI classes.
 biosynthesis spiroketal-amines spiroxamine Low to medium risk. (erg24, erg2) See FRAC SBI
 Guidelines for resistance management G3 KRI fungicides hydroxylanilides fenhexamid Low to
 medium risk. 17 3-keto reductase, (KetoReductase amino-pyrazolinone fenpyrazamine Resistance
 management C4- de-methylation Inhibitors) required. (erg27) (SBI: Class III) G4 (SBI class IV)
 thiocarbamates pyributicarb Resistance not known, 18 squalene-epoxidase allylamines naftifine
 fungicidal and herbicidal in sterol terbinafine activity. biosynthesis Medical fungicides only. (erg1)
 H: H3 Formerly glucopyranosyl reclassified to U18 26 cell antibiotic (validamycin) wall H4
 polyoxins peptidyl pyrimidine polyoxin Resistance known. 19 biosynthesis chitin synthase
 nucleoside Medium risk. Resistance management required. H5 CAA-fungicides cinnamic acid
 amides dimethomorph Resistance known in 40 cellulose synthase (Carboxylic Acid flumorph
Plasmopara viticola but not in Amides) pyrimorph *Phytophthora infestans*. valinamide
 benthiavalicarb Cross resistance between all carbamates iprovalicarb members of the CAA group.
 valifenalate Low to medium risk. mandelic acid amides mandipropamid See FRAC CAA
 Guidelines for resistance management. I: I1 MBI-R isobenzo-furanone fthalide Resistance not
 known. 16.1 melanin reductase in (Melanin pyrrolo-quinolinone pyroquilon synthesis melanin
 Biosynthesis triazolobenzo- tricyclazole in cell biosynthesis Inhibitors - thiazole wall Reductase) I2
 MBI-D cyclopropane- carpropamid Resistance known. 16.2 dehydratase in (Melanin carboxamide
 Medium risk. melanin Biosynthesis carboxamide diclocymet Resistance management biosynthesis
 Inhibitors - propionamide fenoxanil required. Dehydratase) I3 MBI-P trifluoroethyl- tolprocarb
 Resistance not known. 16.3 polyketide synthase (Melanin carbamate Additional activity against in
 melanin Biosynthesis bacteria and fungi through biosynthesis Inhibitors - induction of host plant
 defence Polyketide synthase) P: P 01 benzo- benzo-thiadiazole acibenzolar-S-methyl Resistance
 not known. P 01 host salicylate-related thiadiazole (BTH) plant (BTH) defence P 02
 benzisothiazole benzisothiazole probenazole Resistance not known. P 02 induction salicylate-
 related (also antibacterial and antifungal activity) P 03 thiadiazole- thiadiazole- tiadinil Resistance
 not known. P 03 salicylate-related carboxamide carboxamide isotianil P 04 natural polysaccharides
 laminarin Resistance not known. P 04 polysaccharide compound elicitors P 05 plant extract
 complex mixture, extract from Resistance not known. P 05 anthraquinone ethanol extract
Reynoutria elicitors (anthraquinones, *sachalinensis* resveratrol) (giant knotweed) P 06 microbial
 bacterial *Bacillus mycoides* Resistance not known. P 06 microbial elicitors *Bacillus* spp. isolate J
 fungal cell walls of *Saccharomyces* spp. *Saccharomyces cerevisiae* strain LAS117 P 07
 phosphonates ethyl phosphonates fosetyl-Al Few resistance cases P07 phosphonates phosphorous
 acid reported in few and salts pathogens. Low risk Reclassified from U33 in 2018 P 08 isothiazole
 isothiazolylmethyl dichlobentiazox activates SAR both up- P 08 salicylate-related ether and
 downstream of SA. Resistance not known. U: unknown cyanoacetamide- cyanoacetamide-
 cymoxanil Resistance claims described. 27 Unknown oxime oxime Low to medium risk. mode of
 Resistance management action required. (U numbers formerly phosphonates (FRAC code 33),
 reclassified to P 07 in 2018 not appearing in the unknown phthalamic acids phthalamic acids
 tecloftalam Resistance not known. 34 list (Bactericide) derive unknown benzotriazines
 benzotriazines triazoxide Resistance not known. 35 from unknown benzene- benzene- flusulfamide
 Resistance not known. 36 reclassified sulfonamides sulphonamides fungicides) unknown
 pyridazinones pyridazinones diclomezine Resistance not known. 37 formerly methasulfocarb
 (FRAC code 42), reclassified to M 12 in 2018 unknown phenyl- phenyl-acetamide cyflufenamid
 Resistance in *Sphaerotheca*. U 06 acetamide Resistance management required cell membrane
 guanidines guanidines dodine Resistance known in U 12 disruption *Venturia inaequalis*. (proposed)
 Low to medium risk. Resistance management recommended. unknown thiazolidine cyano-
 methylene- flutianil Resistance in *Sphaerotheca* U 13 thiazolidines and *Podosphaera xanthii*.
 Resistance management required. unknown pyrimidinone- pyrimidinone- ferimzone Resistance not
 known U 14 hydrazones hydrazones (previously C5). complex III: 4-quinolyl- 4-quinolyl-acetates

tebufloquin Not cross resistant to Qol. U 16 cytochrome bc1, acetate Resistance risk unknown but unknown binding assumed to be medium. site (proposed) Resistance management required. Unknown tetrazolyloxime tetrazolyloximes picarbutrazox Resistance not known. U 17 Not cross resistant to PA, Qol, CAA. Unknown glucopyranosyl glucopyranosyl validamycin Resistance not known. U 18 (Inhibition of antibiotic antibiotics Induction of host plant trehalase) defense by trehalose proposed (previously H3). Not Unknown diverse diverse mineral oils, Resistance not known. NC specified organic oils, inorganic salts, material of biological origin M: multi-site inorganic inorganic copper Also applies to organic M 01 Chemicals contact (electrophiles) (different salts) copper complexes with activity inorganic inorganic sulphur generally considered as a low M 02 multi-site (electrophiles) risk group without any signs of activity dithiocarbamates dithio-carbamates amobam resistance developing to the M 03 and relatives and relatives ferbam fungicides. (electrophiles) mancozeb reclassified from U42 in 2018 maneb metiram propineb thiram zinc thiazole zineb ziram phthalimides phthalimides captan M 04 (electrophiles) captafol folpet chloronitriles chloronitriles chlorothalonil M 05 (phthalonitriles) (phthalonitriles) (unspecified mechanism) sulfamides sulfamides dichlofluanid M 06 (electrophiles) tolylfluanid bis-guanidines bis-guanidines guazatine M 07 (membrane iminocadine disruptors, detergents) triazines triazines anilazine M 08 (unspecified mechanism) quinones quinones dithianon M 09 (anthraquinones) (anthraquinones) (electrophiles) quinoxalines quinoxalines chinomethionat/ M 10 (electrophiles) quinomethionate maleimide maleimide fluoroimide M 11 (electrophiles) thiocarbamate thiocarbamate methasulfocarb M 12 (electrophiles) BM: multiple effects on plant extract polypeptide (lectin) extract from the Resistance not known. BM 01 Biologicals ion membrane cotyledons of (previously M12). with transporters; lupine plantlets multiple chelating effects ("BLAD") modes of affects fungal plant extract phenols, extract from Resistance not known. action: spores and germ sesquiterpenes, *Swinglea glutinosa* Plant tubes, triterpenoids, extracts induced plant coumarins defense cell membrane plant extract terpene hydrocarbons, extract from Resistance not known. disruption, cell wall, terpene alcohols and *Melaleuca* (previously F7) induced plant terpene phenols *alternifolia* defense (tea tree oil) mechanisms plant oils (mixtures): eugenol, geraniol, thymol BM: multiple effects microbial fungal *T. atroviride* nomenclature change from BM 02 Biologicals described (strains of living *Trichoderma* spp. strain I-1237 *Gliocladium catenulatum* to with (examples, not all microbes or strain LU132 *Clonostachys rosea* multiple apply to all extract, strain SC1 Resistance not known. modes of biological groups): metabolites) strain SKT-1 *Bacillus amyloliquefaciens* action: competition, strain 77B reclassified from F6, Code 44 Microbial mycoparasitism, *T. asperellum* in 2020 (living antibiosis, strain T34 synonyms for *Bacillus* microbes, membrane strain kd *amyloliquefaciens* are extracts disruption by *T. harzianum* *Bacillus subtilis* or fungicidal strain T-22 and *B. subtilis* var. metabolites) lipopeptides, *T. virens amyloliquefaciens* (previous lytic enzymes, strain G-41 taxonomic classification). induced plant fungal *C. rosea* defence *Clonostachys* spp. strain J1446 strain CR-7 fungal *C. minitans* *Coniothyrium* spp. strain CON/M/91-08 fungal *H. uvarum* *Hanseniaspora* spp. strain BC18Y fungal *T. flavus* *Talaromyces* spp. strain SAY-Y-94-01 fungal *S. cerevisiae* *Saccharomyces* spp. strain LAS02 strain DDSF623 bacterial *B. amyloliquefaciens* *Bacillus* spp. strain QST713 strain FZB24 strain MBI600 strain D747 strain F727 strain AT-332 *B. subtilis* strain AFS032321 strain Y1336 strain HAI-0404 bacterial PHC25279 *Erwinia* spp. (peptide) bacterial *G. cerinus* *Gluconobacter* spp. strain BC18B bacterial *P. chlororaphis* *Pseudomonas* spp. strain AFS009 bacterial *S. griseovirides* *Streptomyces* spp. strain K61 *S. lydicus* strain WYEC108

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Appendix 2

[0198] Shown below is Appendix 2.

TABLE-US-00007 MODE OF ACTION CHEMICAL CLASSIFICATION ACTIVE Inhibition of Acetyl CoA Cyclohexanediones (DIMs) Alloxydim Carboxylase Inhibition of Acetyl CoA Cyclohexanediones (DIMs) Butroxydim Carboxylase Inhibition of Acetyl CoA Cyclohexanediones

(DIMs) Clethodim Carboxylase Inhibition of Acetyl CoA Cyclohexanediones (DIMs) Cloproxydim Carboxylase Inhibition of Acetyl CoA Cyclohexanediones (DIMs) Cycloxydim Carboxylase Inhibition of Acetyl CoA Cyclohexanediones (DIMs) Profoxydim Carboxylase Inhibition of Acetyl CoA Cyclohexanediones (DIMs) Sethoxydim Carboxylase Inhibition of Acetyl CoA Cyclohexanediones (DIMs) Tepraloxym Carboxylase Inhibition of Acetyl CoA Cyclohexanediones (DIMs) Tralkoxydim Carboxylase Inhibition of Acetyl CoA Aryloxyphenoxy-propionates Clodinafop-propargyl Carboxylase (FOPs) Inhibition of Acetyl CoA Aryloxyphenoxy-propionates Clofop Carboxylase (FOPs) Inhibition of Acetyl CoA Aryloxyphenoxy-propionates Cyhalofop-butyl Carboxylase (FOPs) Inhibition of Acetyl CoA Aryloxyphenoxy-propionates Diclofop-methyl Carboxylase (FOPs) Inhibition of Acetyl CoA Aryloxyphenoxy-propionates Fenoxaprop-ethyl Carboxylase (FOPs) Inhibition of Acetyl CoA Aryloxyphenoxy-propionates Fenthiaprop Carboxylase (FOPs) Inhibition of Acetyl CoA Aryloxyphenoxy-propionates Fluazifop-butyl Carboxylase (FOPs) Inhibition of Acetyl CoA Aryloxyphenoxy-propionates Haloxyfop-methyl Carboxylase (FOPs) Inhibition of Acetyl CoA Aryloxyphenoxy-propionates Isoxapyrifop Carboxylase (FOPs) Inhibition of Acetyl CoA Aryloxyphenoxy-propionates Metamifop Carboxylase (FOPs) Inhibition of Acetyl CoA Aryloxyphenoxy-propionates Quizalofop-ethyl Carboxylase (FOPs) Inhibition of Acetyl CoA Phenylpyrazoline Pinoxaden Carboxylase Inhibition of Acetolactate Pyrimidinyl benzoates Bispyribac-sodium Synthase Inhibition of Acetolactate Pyrimidinyl benzoates Pyribenzoxim (prodrug of Synthase bispyribac) Inhibition of Acetolactate Pyrimidinyl benzoates Pyriftalid Synthase Inhibition of Acetolactate Pyrimidinyl benzoates Pyriminobac-methyl Synthase Inhibition of Acetolactate Pyrimidinyl benzoates Pyriothiobac-sodium Synthase Inhibition of Acetolactate Sulfonanilides Pyrimisulfan Synthase Inhibition of Acetolactate Sulfonanilides Triafamone Synthase Inhibition of Acetolactate Triazolopyrimidine - Type 1 Cloransulam-methyl Synthase Inhibition of Acetolactate Triazolopyrimidine - Type 1 Diclosulam Synthase Inhibition of Acetolactate Triazolopyrimidine - Type 1 Florasulam Synthase Inhibition of Acetolactate Triazolopyrimidine - Type 1 Flumetsulam Synthase Inhibition of Acetolactate Triazolopyrimidine - Type 1 Metosulam Synthase Inhibition of Acetolactate Triazolopyrimidine - Type 2 Penoxsulam Synthase Inhibition of Acetolactate Triazolopyrimidine - Type 2 Pyroxsulam Synthase Inhibition of Acetolactate Sulfonylureas Amidosulfuron Synthase Inhibition of Acetolactate Sulfonylureas Azimsulfuron Synthase Inhibition of Acetolactate Sulfonylureas Bensulfuron-methyl Synthase Inhibition of Acetolactate Sulfonylureas Chlorimuron-ethyl Synthase Inhibition of Acetolactate Sulfonylureas Chlorsulfuron Synthase Inhibition of Acetolactate Sulfonylureas Cinosulfuron Synthase Inhibition of Acetolactate Sulfonylureas Cyclosulfamuron Synthase Inhibition of Acetolactate Sulfonylureas Ethametsulfuron-methyl Synthase Inhibition of Acetolactate Sulfonylureas Ethoxysulfuron Synthase Inhibition of Acetolactate Sulfonylureas Flazasulfuron Synthase Inhibition of Acetolactate Sulfonylureas Flucetosulfuron Synthase Inhibition of Acetolactate Sulfonylureas Flupyrsulfuron-methyl-Na Synthase Inhibition of Acetolactate Sulfonylureas Foramsulfuron Synthase Inhibition of Acetolactate Sulfonylureas Halosulfuron-methyl Synthase Inhibition of Acetolactate Sulfonylureas Imazosulfuron Synthase Inhibition of Acetolactate Sulfonylureas Iodosulfuron-methyl-Na Synthase Inhibition of Acetolactate Sulfonylureas Mesosulfuron-methyl Synthase Inhibition of Acetolactate Sulfonylureas Metazosulfuron Synthase Inhibition of Acetolactate Sulfonylureas Metsulfuron-methyl Synthase Inhibition of Acetolactate Sulfonylureas Nicosulfuron Synthase Inhibition of Acetolactate Sulfonylureas Orthosulfamuron Synthase Inhibition of Acetolactate Sulfonylureas Oxasulfuron Synthase Inhibition of Acetolactate Sulfonylureas Primisulfuron-methyl Synthase Inhibition of Acetolactate Sulfonylureas Propyrisulfuron Synthase Inhibition of Acetolactate Sulfonylureas Prosulfuron Synthase Inhibition of Acetolactate Sulfonylureas Pyrazosulfuron-ethyl Synthase Inhibition of Acetolactate Sulfonylureas Rimsulfuron Synthase Inhibition of Acetolactate Sulfonylureas Sulfometuron-methyl Synthase Inhibition of Acetolactate Sulfonylureas Sulfosulfuron Synthase Inhibition of Acetolactate Sulfonylureas Triasulfuron Synthase Inhibition

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Ureas Bromuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Buturon PSII -
Serine 264 Binders Inhibition of Photosynthesis at Ureas Chlorbromuron PSII - Serine 264 Binders

Inhibition of Photosynthesis at Ureas Chlorotoluron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Chloroxuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Difenoxuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Dimefuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Diuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Ethidimuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Fenuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Fluometuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Fluothiuuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Isoproturon PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Isouron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Linuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Metobenzuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Metobromuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Methabenzthiazuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Metoxuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Monolinuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Monuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Neburon PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Parafluron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Siduron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Tebuthiuron PSII - Serine 264 Binders Inhibition of Photosynthesis at Ureas Thiazafluron PSII - Serine 264 Binders Inhibition of Photosynthesis at Amides Chloranocryl = dicryl PSII - Serine 264 Binders Inhibition of Photosynthesis at Amides Pentanochlor PSII - Serine 264 Binders Inhibition of Photosynthesis at Amides Propanil PSII - Serine 264 Binders Inhibition of Photosynthesis at Nitriles Bromofenoxim PSII - Histidine 215 Binders Inhibition of Photosynthesis at Nitriles Bromoxynil PSII - Histidine 215 Binders Inhibition of Photosynthesis at Nitriles Ioxynil PSII - Histidine 215 Binders Inhibition of Photosynthesis at Phenyl-pyridazines Pyridate PSII - Histidine 215 Binders Inhibition of Photosynthesis at Benzothiadiazinone Bentazon PSII - Histidine 215 Binders PS I Electron Diversion Pyridiniums Cyperquat PS I Electron Diversion Pyridiniums Diquat PS I Electron Diversion Pyridiniums Morfamquat PS I Electron Diversion Pyridiniums Paraquat Inhibition of Diphenyl ethers Lactofen Protoporphyrinogen Oxidase Inhibition of Diphenyl ethers Acifluorfen Protoporphyrinogen Oxidase Inhibition of Diphenyl ethers Bifenox Protoporphyrinogen Oxidase Inhibition of Diphenyl ethers Chlornitrofen Protoporphyrinogen Oxidase Inhibition of Diphenyl ethers Fomesafen Protoporphyrinogen Oxidase Inhibition of Diphenyl ethers Fluorodifen Protoporphyrinogen Oxidase Inhibition of Diphenyl ethers Fluoroglycofen-ethyl Protoporphyrinogen Oxidase Inhibition of Diphenyl ethers Fluoronitrofen Protoporphyrinogen Oxidase Inhibition of Diphenyl ethers Nitrofen Protoporphyrinogen Oxidase Inhibition of Diphenyl ethers Oxyfluorfen Protoporphyrinogen Oxidase Inhibition of Diphenyl ethers Chlormethoxyfen Protoporphyrinogen Oxidase Inhibition of Phenylpyrazoles Pyraflufen-ethyl Protoporphyrinogen Oxidase Inhibition of N-Phenyl-oxadiazolones Oxadiargyl Protoporphyrinogen Oxidase Inhibition of N-Phenyl-oxadiazolones Oxadiazon Protoporphyrinogen Oxidase Inhibition of N-Phenyl-triazolinones Azafenidin Protoporphyrinogen Oxidase Inhibition of N-Phenyl-triazolinones Carfentrazone-ethyl Protoporphyrinogen Oxidase Inhibition of N-Phenyl-triazolinones Sulfentrazone Protoporphyrinogen Oxidase Inhibition of N-Phenyl-imides (procide Fluthiacet-methyl Protoporphyrinogen Oxidase active form) Inhibition of N-Phenyl-imides Butafenacil Protoporphyrinogen Oxidase Inhibition of N-Phenyl-imides Saflufenacil Protoporphyrinogen Oxidase Inhibition of N-Phenyl-imides Pentoxazone Protoporphyrinogen Oxidase Inhibition of N-Phenyl-imides Chlorphthalim Protoporphyrinogen Oxidase Inhibition of N-Phenyl-imides Cinidon-ethyl Protoporphyrinogen Oxidase Inhibition of N-Phenyl-imides Flumiclorac-pentyl Protoporphyrinogen Oxidase Inhibition of N-Phenyl-imides Flumioxazin Protoporphyrinogen Oxidase Inhibition of N-Phenyl-imides Flumipropyn Protoporphyrinogen Oxidase Inhibition of N-Phenyl-imides Trifludimoxazin Protoporphyrinogen Oxidase Inhibition of N-Phenyl-imides Tiafenacil Protoporphyrinogen Oxidase Inhibition of Other Pyraclonil

Protoporphyrinogen Oxidase Inhibition of Phytoene Phenyl ethers Beflubenbutamid Desaturase
Inhibition of Phytoene Phenyl ethers Diflufenican Desaturase Inhibition of Phytoene Phenyl ethers
Picolinafen Desaturase Inhibition of Phytoene N-Phenyl heterocycles Flurochloridone Desaturase
Inhibition of Phytoene N-Phenyl heterocycles Norflurazon Desaturase Inhibition of Phytoene
Diphenyl heterocycles Fluridone Desaturase Inhibition of Phytoene Diphenyl heterocycles
Flurtamone Desaturase Inhibition of Hydroxyphenyl Triketones Mesotrione Pyruvate Dioxygenase
Inhibition of Hydroxyphenyl Triketones Sulcotrione Pyruvate Dioxygenase Inhibition of
Hydroxyphenyl Triketones Tembotrione Pyruvate Dioxygenase Inhibition of Hydroxyphenyl
Triketones Tefuryltrione Pyruvate Dioxygenase Inhibition of Hydroxyphenyl Triketones
Bicyclopyrone Pyruvate Dioxygenase Inhibition of Hydroxyphenyl Triketones Fenquinotrione
Pyruvate Dioxygenase Inhibition of Hydroxyphenyl Triketones (procide) Benzobicyclon Pyruvate
Dioxygenase Inhibition of Hydroxyphenyl Pyrazoles (procide) Benzofenap Pyruvate Dioxygenase
Inhibition of Hydroxyphenyl Pyrazoles Pyrasulfotole Pyruvate Dioxygenase Inhibition of
Hydroxyphenyl Pyrazoles Topramezone Pyruvate Dioxygenase Inhibition of Hydroxyphenyl
Pyrazoles (procide) Pyrazolynate Pyruvate Dioxygenase Inhibition of Hydroxyphenyl Pyrazoles
(procide) Pyrazoxyfen Pyruvate Dioxygenase Inhibition of Hydroxyphenyl Pyrazoles Tolpyralate
Pyruvate Dioxygenase Inhibition of Hydroxyphenyl Isoxazoles Isoxaflutole Pyruvate Dioxygenase
Inhibition of Homogentisate Phenoxy pyridazine Cyclopyrimorate Solanesyltransferase Inhibition
of Deoxy-D-Xyulose Isoxazolidinone Clomazone Phosphate Synthase Inhibition of Deoxy-D-
Xyulose Isoxazolidinone Bixlozone Phosphate Synthase Inhibition of Enolpyruvyl Glycine
Glyphosate Shikimate Phosphate Synthase Inhibition of Glutamine Phosphinic acids Glufosinate-
ammonium Synthetase Inhibition of Glutamine Phosphinic acids Bialaphos/bilanafos Synthetase
Inhibition of Dihydropteroate Carbamate Asulam Synthase Inhibition of Microtubule
Dinitroanilines Benefin = Assembly benfluralin Inhibition of Microtubule Dinitroanilines Butralin
Assembly Inhibition of Microtubule Dinitroanilines Dinitramine Assembly Inhibition of
Microtubule Dinitroanilines Ethalfluralin Assembly Inhibition of Microtubule Dinitroanilines
Fluchloralin Assembly Inhibition of Microtubule Dinitroanilines Isopropalin Assembly Inhibition
of Microtubule Dinitroanilines Nitalin Assembly Inhibition of Microtubule Dinitroanilines
Prodiamine Assembly Inhibition of Microtubule Dinitroanilines Profluralin Assembly Inhibition of
Microtubule Dinitroanilines Oryzalin Assembly Inhibition of Microtubule Dinitroanilines
Pendimethalin Assembly Inhibition of Microtubule Dinitroanilines Trifluralin Assembly Inhibition
of Microtubule Pyridines Dithiopyr Assembly Inhibition of Microtubule Pyridines Thiazopyr
Assembly Inhibition of Microtubule Phosphoroamidates Butamifos Assembly Inhibition of
Microtubule Phosphoroamidates DMPA Assembly Inhibition of Microtubule Benzoic acid
Chlorthal-dimethyl = Assembly DCPA Inhibition of Microtubule Benzamides Propyzamide =
Assembly pronamide Inhibition of Microtubule Carbamates Barban Organization Inhibition of
Microtubule Carbamates Carbetamide Organization Inhibition of Microtubule Carbamates
Chlorbufam Organization Inhibition of Microtubule Carbamates Chlorpropham Organization
Inhibition of Microtubule Carbamates Propham Organization Inhibition of Microtubule
Carbamates Swep Organization Inhibition of Cellulose Triazolocarboxamide Flupoxam Synthesis
Inhibition of Cellulose Benzamides Isoxaben Synthesis Inhibition of Cellulose Alkylazines
Triaziflam Synthesis Inhibition of Cellulose Alkylazines Indaziflam Synthesis Inhibition of
Cellulose Nitriles Dichlobenil Synthesis Inhibition of Cellulose Nitriles Chlorthiamid Synthesis
Uncouplers Dinitrophenols Dinosam Uncouplers Dinitrophenols Dinoseb Uncouplers
Dinitrophenols DNOC Uncouplers Dinitrophenols Dinoterb Uncouplers Dinitrophenols Etinofen
Uncouplers Dinitrophenols Medinoterb Inhibition of Very Long-Chain Azolyl-carboxamides
Cafenstrole Fatty Acid Synthesis Inhibition of Very Long-Chain Azolyl-carboxamides
Fentrazamide Fatty Acid Synthesis Inhibition of Very Long-Chain Azolyl-carboxamides
Ipfencarbazone Fatty Acid Synthesis Inhibition of Very Long-Chain α -Thioacetamides Anilofos
Fatty Acid Synthesis Inhibition of Very Long-Chain α -Thioacetamides Piperophos Fatty Acid

Synthesis Inhibition of Very Long-Chain Isoxazoles Pyroxasulfone Fatty Acid Synthesis Inhibition of Very Long-Chain Isoxazoles Fenoxasulfone Fatty Acid Synthesis Inhibition of Very Long-Chain Oxiranes Indanofan Fatty Acid Synthesis Inhibition of Very Long-Chain Oxiranes Tridiphane Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Acetochlor Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Alachlor Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Allidochlor = Fatty Acid Synthesis CDAA Inhibition of Very Long-Chain α -Chloroacetamides Butachlor Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Butenachlor Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Delachlor Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Diethatyl-ethyl Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Dimethachlor Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Dimethenamid Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Metazachlor Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Metolachlor Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Pethoxamid Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Pretilachlor Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Propachlor Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Propisochlor Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Prynachlor Fatty Acid Synthesis Inhibition of Very Long-Chain α -Chloroacetamides Thenylchlor Fatty Acid Synthesis Inhibition of Very Long-Chain α -Oxyacetamides Mefenacet Fatty Acid Synthesis Inhibition of Very Long-Chain α -Oxyacetamides Flufenacet Fatty Acid Synthesis Inhibition of Very Long-Chain Thiocarbamates Butylate Fatty Acid Synthesis Inhibition of Very Long-Chain Thiocarbamates Cycloate Fatty Acid Synthesis Inhibition of Very Long-Chain Thiocarbamates Dimepiperate Fatty Acid Synthesis Inhibition of Very Long-Chain Thiocarbamates EPTC Fatty Acid Synthesis Inhibition of Very Long-Chain Thiocarbamates Esprocarb Fatty Acid Synthesis Inhibition of Very Long-Chain Thiocarbamates Molinate Fatty Acid Synthesis Inhibition of Very Long-Chain Thiocarbamates Orbencarb Fatty Acid Synthesis Inhibition of Very Long-Chain Thiocarbamates Pebulate Fatty Acid Synthesis Inhibition of Very Long-Chain Thiocarbamates Prosulfocarb Fatty Acid Synthesis Inhibition of Very Long-Chain Thiocarbamates Thiobencarb Fatty Acid Synthesis (=Benthicarb) Inhibition of Very Long-Chain Thiocarbamates Tiocarbazil Fatty Acid Synthesis Inhibition of Very Long-Chain Thiocarbamates Tri-allate Fatty Acid Synthesis Inhibition of Very Long-Chain Thiocarbamates Vernolate Fatty Acid Synthesis Inhibition of Very Long-Chain Benzofurans Benfuresate Fatty Acid Synthesis Inhibition of Very Long-Chain Benzofurans Ethofumesate Fatty Acid Synthesis Auxin Mimics Pyridine-carboxylates Picloram Auxin Mimics Pyridine-carboxylates Clopyralid Auxin Mimics Pyridine-carboxylates Aminopyralid Auxin Mimics Pyridine-carboxylates Halauxifen Auxin Mimics Pyridine-carboxylates Flurpyrauxifen Auxin Mimics Pyridyloxy-carboxylates Triclopyr Auxin Mimics Pyridyloxy-carboxylates Fluroxypyr Auxin Mimics Phenoxy-carboxylates 2,4,5-T Auxin Mimics Phenoxy-carboxylates 2,4-D Auxin Mimics Phenoxy-carboxylates 2,4-DB Auxin Mimics Phenoxy-carboxylates Clomeprop Auxin Mimics Phenoxy-carboxylates Dichlorprop Auxin Mimics Phenoxy-carboxylates Fenoprop Auxin Mimics Phenoxy-carboxylates Mecoprop Auxin Mimics Phenoxy-carboxylates MCPA Auxin Mimics Phenoxy-carboxylates MCPB Auxin Mimics Benzoates Dicamba Auxin Mimics Benzoates Chloramben Auxin Mimics Benzoates TBA Auxin Mimics Quinoline-carboxylates Quinclorac Auxin Mimics Quinoline-carboxylates Quinmerac Auxin Mimics Pyrimidine-carboxylates Aminocyclopyrachlor Auxin Mimics Other Benazolin-ethyl Auxin Mimics Phenyl carboxylates Chlorfenac = fenac Auxin Mimics Phenyl carboxylates Chlorfenprop Auxin Transport Inhibitor Aryl-carboxylates Naptalam Auxin Transport Inhibitor Aryl-carboxylates Diflufenzopyr-sodium Inhibition of Fatty Acid Benzyl ether Cinmethylin Thioesterase Inhibition of Fatty Acid Benzyl ether Methiozolin Thioesterase Inhibition of Serine-Threonine Other Endothal Protein Phosphatase Inhibition of Solanesyl Diphenyl ether Aclonifen

Diphosphate Synthase Inhibition of Lycopene Triazole Amitrole Cyclyase Unknown Bromobutide
 Unknown Cumyluron Unknown Difenzoquat Unknown DSMA Unknown Dymron = Daimuron
 Unknown Etobenzanid Unknown Arylamino propionic acid Flamprop-m Unknown Fosamine
 Unknown Methyldymron Unknown Monalide Unknown MSMA Unknown Oleic acid Unknown
 Oxaziclomefone Unknown Pelargonic acid Unknown Pyributicarb Unknown Quinoclamine
 Unknown Acetamides Diphenamid Unknown Acetamides Naproanilide Unknown Acetamides
 Napropamide Unknown Benzamide Tebutam Unknown Phosphorodithioate Bensulide Unknown
 Chlorocarbonic acids Dalapon Unknown Chlorocarbonic acids Flupropanate Unknown
 Chlorocarbonic acids TCA Unknown Trifluoromethanesulfonanilides Mefluidide Unknown
 Trifluoromethanesulfonanilides Perfluidone Unknown CAMA Unknown Cacodylic acid

Appendix 3

[0199] Shown below is Appendix 3.

TABLE-US-00008 Sub-group, class or Main Group and Primary Site of exemplifying Active
 Action Ingredient Active Ingredients 1 1A Alanycarb, Aldicarb, Bendiocarb, Benfuracarb,
 Butocarboxim, Acetylcholinesterase (AChE) Carbamates Butoxycarboxim, Carbaryl, Carbofuran,
 Carbosulfan, inhibitors Ethiofencarb, Fenobucarb, Formetanate, Furathiocarb, Nerve action
 Isoprocab, Methiocarb, Methomyl, Metolcarb, Oxamyl, {Strong evidence that action at this
 Pirimicarb, Propoxur, Thiodicarb, Thiofanox, protein is responsible for insecticidal Triazamate,
 Trimethacarb, XMC, Xylcarb effects} 1B Acephate, Azamethiphos, Azinphos-ethyl, Azinphos-
 Organophosphates methyl, Cadusafos, Chlorethoxyfos, Chlorfenvinphos, Chlormephos,
 Chlorpyrifos, Chlorpyrifos-methyl, Coumaphos, Cyanophos, Demeton-S-methyl, Diazinon,
 Dichlorvos/ DDVP, Dicrotophos, Dimethoate, Dimethylvinphos, Disulfoton, EPN, Ethion,
 Ethoprophos, Famphur, Fenamiphos, Fenitrothion, Fenthion, Fosthiazate, Heptenophos, Imicyafos,
 Isofenphos, Isopropyl O-(methoxyaminothio- phosphoryl) salicylate, Isoxathion, Malathion,
 Mecarbam, Methamidophos, Methidathion, Mevinphos, Monocrotophos, Naled, Omethoate,
 Oxydemeton-methyl, Parathion, Parathion- methyl, Phenthoate, Phorate, Phosalone, Phosmet,
 Phosphamidon, Phoxim, Pirimiphos- methyl, Profenofos, Propetamphos, Prothiofos, Pyraclofos,
 Pyridaphenthion, Quinalphos, Sulfotep, Tebupirimfos, Temephos, Terbufos, Tetrachlorvinphos,
 Thiometon, Triazophos, Trichlorfon, Vamidothion 2 2A Chlordane, Endosulfan GABA-gated
 chloride channel blockers Cyclodiene Nerve action Organochlorines {Strong evidence that action
 at this 2B Ethiprole, Fipronil protein is responsible for insecticidal Phenylpyrazoles (Fiproles)
 effects} 3 3A Acrinathrin, Allethrin, d-cis-trans Allethrin, d- trans Allethrin, Sodium channel
 modulators Pyrethroids Pyrethrins Bifenthrin, Bioallethrin, Bioallethrin S-cyclopentenyl isomer,
 Nerve action Bioresmethrin, Cycloprothrin, Cyfluthrin, beta- Cyfluthrin, {Strong evidence that
 action at this Cyhalothrin, lambda-Cyhalothrin, gamma-Cyhalothrin, protein is responsible for
 insecticidal Cypermethrin, alpha- Cypermethrin, beta-Cypermethrin, effects} theta- cypermethrin,
 zeta-Cypermethrin, Cyphenothrin, (1R)-trans- isomers], Deltamethrin, Empenthrin (EZ)- (1R)-
 isomers], Esfenvalerate, Etofenprox, Fenpropathrin, Fenvalerate, Flucythrinate, Flumethrin, tau-
 Fluvalinate, Halfenprox, Imiprothrin, Kadethrin, Permethrin, Phenothrin [(1R)-trans- isomer],
 Prallethrin, Pyrethrins (pyrethrum), Resmethrin, Silafluofen, Tefluthrin, Tetramethrin, Tetramethrin
 [(1R)-isomers], Tralomethrin, Transfluthrin, 3B DDT DDT Methoxychlor Methoxychlor 4 4A
 Acetamiprid, Clothianidin, Dinotefuran, Nicotinic acetylcholine receptor Neonicotinoids
 Imidacloprid, Nitenpyram, Thiacloprid, Thiamethoxam, (nAChR) competitive modulators 4B
 Nicotine Nerve action Nicotine {Strong evidence that action at one or 4C Sulfoxaflor more of this
 class of protein is Sulfoximines responsible for insecticidal effects} 4D Flupyradifurone
 Butenolides 4E Triflumezopyrim Mesoionics 4F Flupyrimin Pyridylidenes 5 Spinosyns
 Spinetoram, Spinosad Nicotinic acetylcholine receptor (nAChR) allosteric modulators - Site I
 Nerve action {Strong evidence that action at one or more of this class of protein is responsible for
 insecticidal effects} 6 Avermectins, Abamectin, Emamectin benzoate, Lepimectin, Milbemectin
 Glutamate-gated chloride Milbemycins channel (GluCl) allosteric modulators Nerve and muscle

action {Strong evidence that action at one or more of this class of protein is responsible for insecticidal effects} 7 7A Hydroprene, Kinoprene, Methoprene Juvenile hormone mimics Juvenile hormone Growth regulation analogues {Target protein responsible for biological 7B Fenoxycarb activity is unknown, or uncharacterized} Fenoxycarb 7C Pyriproxyfen Pyriproxyfen 8 * 8A Methyl bromide and other alkyl halides Miscellaneous non-specific (multi- Alkyl halides site) inhibitors 8B Chloropicrin Chloropicrin 8C Cryolite (Sodium aluminum fluoride), Sulfuryl fluoride Fluorides 8D Borax, Boric acid, Disodium octaborate, Sodium borate, Borates Sodium metaborate 8E Tartar emetic Tartar emetic 8F Dazomet, Metam Methyl isothiocyanate generators 9 9B Pymetrozine, Pyriproxyfen Chordotonal organ TRPV Pyridine azomethine channel modulators Nerve action derivatives {Strong evidence that action at one or 9D Afidopyropen more of this class of proteins is Pyropenes responsible for insecticidal effects} 10 10A Clofentezine, Diflovidazin, Hexythiazox Mite growth inhibitors affecting Clofentezine Diflovidazin CHS1 Hexythiazox Growth regulation 10B Etoxazole {Strong evidence that action at one or Etoxazole more of this class of proteins is responsible for insecticidal effects} 11 11A *Bacillus thuringiensis* subsp. *israelensis* *Bacillus* Microbial disruptors of insect midgut *Bacillus thuringiensis* and *thuringiensis* subsp. *aizawai* *Bacillus thuringiensis* membranes the insecticidal proteins subsp. *kurstaki* *Bacillus thuringiensis* subsp. (Includes transgenic crops expressing they produce *tenebrionis* *Bacillus thuringiensis* toxins, however B.t. crop proteins: (* Please see footnote) Cry1Ab, Cry1Ac, specific guidance for resistance Cry1Fa, Cry1A.105, Cry2Ab, Vip3A, mCry3A, Cry3Ab, management of transgenic crops is not Cry3Bb, Cry34Ab1/Cry35Ab1 based on rotation of modes of action) 11B *Bacillus sphaericus* *Bacillus sphaericus* 12 12A Diafenthiuron Inhibitors of mitochondrial ATP Diafenthiuron synthase 12B Azocyclotin, Cyhexatin, Fenbutatin oxide Energy metabolism Organotin miticides {Compounds affect the function of this 12C Propargite protein, but it is not clear that this is what Propargite leads to biological activity} 12D Tetradifon Tetradifon 13 * Pyrroles Dinitrophenols Chlorfenapyr DNOC Uncouplers of oxidative Sulfluramid Sulfluramid phosphorylation via disruption of the proton gradient Energy metabolism 14 Nereistoxin analogues Bensultap, Cartap hydrochloride, Thiocyclam, Nicotinic acetylcholine receptor Thiosultap-sodium (nAChR) channel blockers Nerve action {Compounds affect the function of this protein, but it is not clear that this is what leads to biological activity} 15 Benzoylureas Bistrifluron, Chlorfluazuron, Diflubenzuron, Flucyclohexuron, Inhibitors of chitin biosynthesis Flufenoxuron, Hexaflumuron, Lufenuron, Novaluron, affecting CHS1 Noviflumuron, Teflubenzuron, Triflumuron Growth regulation {Strong evidence that action at one or more of this class of proteins is responsible for insecticidal effects} 16 Buprofezin Buprofezin Inhibitors of chitin biosynthesis, type 1 Growth regulation {Target protein responsible for biological activity is unknown, or uncharacterized} 17 Cyromazine Cyromazine Moulting disruptors, Dipteran Growth regulation {Target protein responsible for biological activity is unknown, or uncharacterized} 18 Diacylhydrazines Chromafenozide, Halofenozide, Methoxyfenozide, Ecdysone receptor agonists Tebufenozide Growth regulation {Strong evidence that action at this protein is responsible for insecticidal effects} 19 Amitraz Amitraz Octopamine receptor agonists Nerve action {Good evidence that action at one or more of this class of protein is responsible for insecticidal effects} 20 20A Hydramethylnon Mitochondrial complex III electron Hydramethylnon transport inhibitors - Qo site 20B Acequinocyl Energy metabolism Acequinocyl {Good evidence that action at this 20C Fluacrypyrim protein complex is responsible for Fluacrypyrim insecticidal effects} 20D Bifenazate Bifenazate 21 21A Fenazaquin, Fenpyroximate, Pyridaben, Pyrimidifen, Mitochondrial complex I electron METI acaricides and Tebufenpyrad, Tolfenpyrad transport inhibitors insecticides Energy metabolism 21B Rotenone (Derris) Rotenone {Good evidence that action at this protein complex is responsible for insecticidal effects} 22 22A Indoxacarb Voltage-dependent sodium Oxadiazines channel blockers 22B Metaflumizone Nerve action Semicarbazones {Good evidence that action at this protein complex is responsible for insecticidal effects} 23 Tetronic and Tetramic acid Spirodiclofen, Spiromesifen, Spiropidion, Spirotetramat Inhibitors of acetyl CoA carboxylase

derivatives Lipid synthesis, growth regulation {Good evidence that action at this protein is responsible for insecticidal effects} 24 24A Aluminium phosphide, Calcium phosphide, Phosphine, Zinc Mitochondrial complex IV electron Phosphides phosphide transport inhibitors 24B Calcium cyanide, Potassium cyanide, Sodium cyanide Energy metabolism Cyanides {Good evidence that action at this protein complex is responsible for insecticidal effects} 25 25A Cyenopyrafen, Cyflumetofen Mitochondrial complex II electron Beta-ketonitrile transport inhibitors derivatives Energy metabolism 25B Pyflubumide {Good evidence that action at this Carboxanilides protein complex is responsible for insecticidal effects} 28 Diamides Chlorantraniliprole, Cyantraniliprole, Cyclaniliprole Ryanodine receptor Flubendiamide, Tetraniliprole modulators Nerve and muscle action {Strong evidence that action at this protein complex is responsible for insecticidal effects} 29 Flonicamid Flonicamid Chordotonal organ modulators - undefined target site Nerve action (Modulation of chordotonal organ function has been clearly demonstrated, but the specific target protein(s) responsible for biological activity are distinct from Group 9 and remain undefined) 30 Meta-diamides Isoxazolines Broflanilide GABA-gated chloride channel allosteric Fluxametamide, Isocyloseram modulators Nerve action {Strong evidence that action at this protein complex is responsible for insecticidal effects} 31 Granuloviruses (GVs) *Cydia pomonella* GV Baculoviruses *Thaumatotibia leucotreta* GV Host-specific occluded Nucleopolyhedroviruses *Anticarsia gemmatilis* MNPV pathogenic viruses (NPVs) *Helicoverpa armigera* NPV (Midgut epithelial columnar cell membrane target site - undefined) 32 GS-omega/kappa GS-omega/kappa HXTX-Hv1a peptide Nicotinic Acetylcholine Receptor HXTX-Hv1a peptide (nAChR) Allosteric Modulators - Site II Nerve action {Strong evidence that action at one or more of this class of protein is responsible for insecticidal effects} 33 Acynonapyr Acynonapyr Calcium-activated potassium channel (KCa2) modulators Nerve action {Strong evidence that action at this protein is responsible for insecticidal effects} 34 Flometoquin Flometoquin Mitochondrial complex III electron transport inhibitors - Qi site Energy metabolism {Modulation of this protein complex has been clearly demonstrated and the specific target site responsible for biological activity is distinct from Group 20} UN* Azadirachtin Azadirachtin Compounds of unknown or uncertain Benzoximate Benzoximate MoA Benzpyrimoxan Benzpyrimoxan {Target protein responsible for biological Bromopropylate Bromopropylate activity is unknown, or uncharacterized} Chinomethionat Chinomethionat Dicofol Dicofol Lime sulfur Lime sulfur Mancozeb Mancozeb Pyridalyl Pyridalyl Sulfur Sulfur UNB* *Burkholderia* spp Bacterial agents (non-Bt) of unknown or *Wolbachia pipientis* (Zap) uncertain MoA {Target protein responsible for biological activity is unknown or uncharacterized} UNE* *Chenopodium ambrosioides* near *ambrosioides* Botanical essence including extract synthetic, extracts and unrefined Fatty acid monoesters with glycerol or oils with unknown or uncertain MoA propanediol Neem oil {Target protein responsible for biological activity is unknown, or uncharacterized} UNF* *Beauveria bassiana* strains Fungal agents of unknown or uncertain *Metarhizium anisopliae* strain F52 MoA *Paecilomyces fumosoroseus* Apopka strain 97 {Target protein responsible for biological activity is unknown, or uncharacterized} UNM* Diatomaceous earth Non-specific mechanical and physical Mineral oil disruptors {Target protein responsible for biological activity is unknown, or uncharacterized}

[0200] Notwithstanding the appended claims, aspects of the present disclosure are further provided in the following numbered clauses: [0201] 1. A method for inhibiting apyrase, comprising contacting the apyrase with a compound of formula (I):

##STR00132## [0202] wherein: [0203] G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R.sup.a and R.sup.b; [0204] Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C.sub.1-6 alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R.sup.a and R.sup.b; [0205] R.sup.1 is hydrogen or C.sub.1-6 alkyl; [0206] or Z and R.sup.1 together form a 5-, 6-, or 7-membered ring selected from heterocyclalkyl optionally substituted with 1, 2 or 3 groups selected from R.sup.a and R.sup.b; [0207] each R.sup.a is independently

selected from the group consisting of C.sub.1-6 alkyl, C.sub.3-8 cycloalkyl, C.sub.5-10 aryl, C.sub.6-10 arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclylalkyl, 4-11 membered heterocyclylalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl; [0208] each R.sup.a is optionally substituted with one or more groups selected from R.sup.b and R.sup.e; [0209] each R.sup.b is independently selected from the group consisting of =O, —OR.sup.d, —OCF.sub.3, —OCF.sub.2H=S, —SR.sup.d, =NR.sup.d, =NOR.sup.d, —NR.sup.cR.sup.c, halogen, —CF.sub.3, —CF.sub.2H, —CN, —NO.sub.2, =N.sub.2, —N.sub.3, —S(O)R.sup.d, —S(O).sub.2R.sup.d, —S(O).sub.2OR.sup.d, —S(O)NR.sup.cR.sup.c, —S(O).sub.2NR.sup.cR.sup.c, —NHS(O).sub.2R.sup.d, —OS(O)R.sup.d, —OS(O).sub.2R.sup.d, —OS(O).sub.2OR.sup.d, —OS(O).sub.2NR.sup.cR.sup.c, —C(O)R.sup.d, —C(O)OR.sup.d, —C(O)NR.sup.cR.sup.c, —C(NH)NR.sup.cR.sup.c, —C(NR.sup.a)NR.sup.cR.sup.c, —C(NOH)R.sup.a, —C(NOH)NR.sup.cR.sup.c, —OC(O)R.sup.d, —OC(O)OR.sup.d, —OC(O)NR.sup.cR.sup.c, —OC(NH)NR.sup.cR.sup.c, —OC(NR.sup.a)NR.sup.cR.sup.c, —[NHC(O)].sub.nR.sup.d, —[NR.sup.aC(O)].sub.nR.sup.d, [NHC(O)].sub.nOR.sup.d, —[NR.sup.aC(O)].sub.nOR.sup.d, —[NHC(O)].sub.nNR.sup.cR.sup.c, —[NR.sup.aC(O)].sub.nNR.sup.cR.sup.c, —NHSO.sub.2R.sup.d, —[NHC(NH)].sub.nNR.sup.cR.sup.c and —[NR.sup.aC(NR.sup.a)].sub.nNR.sup.cR.sup.c; [0210] each R.sup.c is independently hydrogen, R.sup.3, or, alternatively, two R.sup.e are taken together with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclylalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of the same or different group selected from R.sup.a, =O, halogen, and R.sup.e; [0211] each R.sup.d is independently C.sub.1-6 alkyl or C.sub.3-8 cycloalkyl; [0212] each R.sup.e is independently C.sub.1-6 alkyl, C.sub.3-8 cycloalkyl, —C(O)R.sup.d, and —C(O)OR.sup.d; [0213] provided that the compound does not have the formula

##STR00133## [0214] 2. The method of clause 1, wherein G comprises a six-membered aryl or six-membered heteroaryl ring. [0215] 3. The method of clause 2, wherein G is an optionally substituted phenyl. [0216] 4. The method of clause 3, wherein G is substituted phenyl. [0217] 5. The method of clause 4, wherein the compound has formula (Ia):

##STR00134## [0218] 6. The method of clause 4, wherein the compound has formula (Ib):

##STR00135## [0219] wherein Ar is optionally substituted aryl or heteroaryl. [0220] 7. The method of clause 6, wherein Ar in formula (Ib) is unsubstituted, substituted with an R.sup.a group of C.sub.1-6 alkyl, or substituted with an R.sup.b group of halogen. [0221] 8. The method of clause 2, wherein G is an optionally substituted biphenyl group. [0222] 9. The method of clause 8, wherein the compound has the formula (Ib1):

##STR00136## [0223] 10. The method of clause 1, wherein G is optionally substituted pyrimidinyl. [0224] 11. The method of clause 10, wherein the compound has formula (Ic):

##STR00137## [0225] wherein Ar is optionally substituted aryl or heteroaryl. [0226] 12. The method of clause 11, wherein Ar in formula (Ic) is unsubstituted, substituted with an R.sup.a group of C.sub.1-6 alkyl, or substituted with an R.sup.b group of halogen. [0227] 13. The method of any one of clauses 1-12, wherein G and its substituents have 6 to 20 non-hydrogen atoms. [0228] 14. The method of any one of clauses 1-13, wherein Z is substituted phenyl. [0229] 15. The method of clause 14, wherein Z is phenyl substituted with an optionally substituted aryl or heteroaryl group. [0230] 16. The method of clause 15, wherein Z is phenyl substituted with an optionally substituted alkyl group. [0231] 17. The method of clause 16, wherein Z is phenyl substituted with an R.sup.b group. [0232] 18. The method of clause 17, wherein the compound has the formula (Id):

##STR00138## [0233] 19. The method of clause 17, wherein the compound has the formula (Ie):

##STR00139## [0234] 20. The method of clause 17, wherein Z is phenyl substituted with an R.sup.b group of halogen or —CF.sub.3. [0235] 21. The method of any one of clauses 1-13, wherein Z is optionally substituted monocyclic heteroaryl. [0236] 22. The method of clause 21,

wherein Z is optionally substituted pyridyl. [0237] 23. The method of clause 21, wherein Z is optionally substituted pyrimidyl. [0238] 24. The method of any one of clauses 1-13, wherein Z and R.sup.1 together form a 5-membered or 6-membered heterocyclylalkyl ring. [0239] 25. The method of any one of clauses 1-13, wherein Z is selected from the group consisting of

##STR00140## ##STR00141## [0240] 26. The method of any one of clauses 1-25, wherein Z and its substituents have 6 to 20 non-hydrogen atoms. [0241] 27. The method of any one of clauses 1-26, wherein contacting the apyrase comprises treating a crop with the compound. [0242] 28. The method of clause 27, further comprising treating the crop with a pesticide. [0243] 29 The method of clause 28, wherein the pesticide is selected from acaricides, fungicides, herbicides, insecticides, molluscicides, nematocides, or a combination thereof. [0244] 30. The method of clause 29, wherein the pesticide comprises a fungicide. [0245] 31. The method of clause 28, wherein the pesticide comprises an herbicide. [0246] 32. The method of clause 31, wherein the herbicide comprises glyphosate. [0247] 33. The method of clause 29, further comprising treating the crop with a fungicide selected from selected from benzimidazoles, dicarboximides, phenylpyrroles, anilinopyrimidines, hydroxyanilides, carboxamides, phenyl amides, phosphonates, cinnamic acids, oxysterol binding protein inhibitors (OSBPI), triazole carboxamides, cymoxanil, carbamates, benzamides, demethylation inhibiting piperazines, demethylation inhibiting pyrimidines, demethylation inhibiting azoles, including imidazoles and triazoles, such as cyproconazole, difenoconazole, fenbuconazole, flutriafol, mefentrifluconazole, metconazole, ipconazole, prothioconazole, tebuconazole, tetraconazole, triadimefon, triadimenol, triticonazole, morpholines, cyflufenamid, metrafenone, pyriofenone, strobilurins, copper ammonium complex, copper hydroxide, copper oxide, copper oxychloride, copper sulfate, sulfur, lime sulfur, ethylenebisdithiocarbamates, aromatic hydrocarbons, phthalimides, guanidines, polyoxins, fluazinam, thiazolidines and combinations thereof. [0248] 34. A composition, comprising [0249] a fungicide; [0250] a phytologically acceptable carrier; and [0251] a compound of formula (I):

##STR00142## [0252] wherein: [0253] G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R.sup.a and R.sup.b; [0254] Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C.sub.1-6 alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R.sup.a and R.sup.b; [0255] R.sup.1 is hydrogen or C.sub.1-6 alkyl; [0256] or Z and R.sup.1 together form a 5-, 6-, or 7-membered ring selected from heterocyclylalkyl optionally substituted with 1, 2 or 3 groups selected from R.sup.a and R.sup.b; [0257] each R.sup.a is independently selected from the group consisting of C.sub.1-6 alkyl, C.sub.3-8 cycloalkyl, C.sub.5-C.sub.10 aryl, C.sub.6-C.sub.10 arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclylalkyl, 4-11 membered heterocyclylalkylalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl; [0258] each R.sup.a is optionally substituted with one or more groups selected from R.sup.b and R.sup.e; [0259] each R.sup.b is independently selected from the group consisting of =O, —OR.sup.d, —OCF.sub.3, —OCF.sub.2H=S, —SR.sup.d, =NR.sup.d, =NOR.sup.d, —NR.sup.cR.sup.c, halogen, —CF.sub.3, —CF.sub.2H, —CN, —NO.sub.2, =N.sub.2, —N.sub.3, —S(O)R.sup.d, —S(O).sub.2R.sup.d, —S(O).sub.2OR.sup.d, —S(O)NR.sup.cR.sup.c, —S(O).sub.2NR.sup.cR.sup.c, —NHS(O).sub.2R.sup.d, —OS(O)R.sup.d, —OS(O).sub.2R.sup.d, —OS(O).sub.2OR.sup.d, —OS(O).sub.2NR.sup.cR.sup.c, —C(O)R.sup.d, —C(O)OR.sup.d, —C(O)N R.sup.cR.sup.c, —C(NH)NR.sup.cR.sup.c, —C(NR.sup.a)NR.sup.cR.sup.c, —C(NOH)R.sup.a, —C(NOH)NR.sup.cR.sup.c, —OC(O)R.sup.d, —OC(O)OR.sup.d, —OC(O)NR.sup.cR.sup.c, —OC(NH)NR.sup.cR.sup.c, —OC(NR.sup.a)NR.sup.cR.sup.c, —[NHC(O)].sub.nR.sup.d, —[NR.sup.aC(O)].sub.nR.sup.d, —[NHC(O)].sub.nOR 4, —[NR.sup.aC(O)].sub.nOR.sup.d, —[NHC(O)].sub.nNR.sup.cR.sup.c, —[NR.sup.aC(O)].sub.nNR.sup.cR.sup.c, —NHSO.sub.2R.sup.d, —[NHC(NH)].sub.nNR.sup.cR.sup.c and —[NR.sup.aC(NR.sup.a)].sub.nNR.sup.cR.sup.c; [0260] each R.sup.c is independently hydrogen, R.sup.a, or, alternatively, two R.sup.c are taken together

with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of the same or different group selected from R.sup.a, =O, halogen, and R.sup.e; [0261] each R.sup.d is independently C.sub.1-6 alkyl or C.sub.3-8 cycloalkyl; and [0262] each R.sup.e is independently C.sub.1-6 alkyl, C.sub.3-8 cycloalkyl, —C(O)R.sup.d, and —C(O)OR.sup.d. [0263] 35. The composition of clause 34, wherein the composition comprises from about 1 to about 80 weight percent of the compound. [0264] 36. The composition of clause 35, wherein the composition is a suspension formulation. [0265] 37. The composition of clause 36, wherein the composition comprises from about 1 to about 50 weight percent of the compound. [0266] 38. The composition of clause 37, further comprising sodium polycarboxylate. [0267] 39. The composition of clause 34, further comprising biocides. [0268] 40. The composition of clause 35, further comprising organosilicone antifoam emulsion. [0269] 41. The composition of clause 38, wherein the composition is a wettable powder. [0270] 42. The composition of clause 34, wherein the composition is an emulsifiable concentrate. [0271] 43. The composition of clause 40, further comprising tristyrylphenol ethoxylates. [0272] 44. The composition of clause 34, wherein the composition is an oil dispersible concentrate. [0273] 45. A pesticidal composition, comprising [0274] a pesticide; [0275] a phytologically acceptable carrier; and [0276] a compound of formula (I):

##STR00143## [0277] wherein: [0278] G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R.sup.a and R.sup.b; [0279] Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C.sub.1-6 alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R.sup.a and R.sup.b; [0280] R.sup.1 is hydrogen or C.sub.1-6 alkyl; [0281] or Z and R.sup.1 together form a 5-, 6-, or 7-membered ring selected from heterocyclalkyl optionally substituted with 1, 2 or 3 groups selected from R.sup.a and R.sup.b; [0282] each R.sup.a is independently selected from the group consisting of C.sub.1-6 alkyl, C.sub.3-8 cycloalkyl, C.sub.5-C.sub.10 aryl, C.sub.6-C.sub.10 arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclalkyl, 4-11 membered heterocyclalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl; [0283] each R.sup.a is optionally substituted with one or more groups selected from R.sup.b and R.sup.e; [0284] each R.sup.b is independently selected from the group consisting of =O, —OR.sup.d, —OCF.sub.3, —OCF.sub.2H=S, —SR.sup.d, =NR.sup.d, =NOR.sup.d, —NR.sup.cR.sup.c, halogen, —CF.sub.3, —CF.sub.2H, —CN, —NO.sub.2, =N.sub.2, —N.sub.3, —S(O)R.sup.d, —S(O).sub.2R.sup.d, —S(O).sub.2OR.sup.d, —S(O)NR.sup.cR.sup.c, —S(O).sub.2NR.sup.cR.sup.c, —NHS(O).sub.2R.sup.d, —OS(O)R.sup.d, —OS(O).sub.2R.sup.d, —OS(O).sub.2OR.sup.d, —OS(O).sub.2NR.sup.cR.sup.c, —C(O)R.sup.d, —C(O)OR.sup.d, —C(O)N R.sup.cR.sup.c, —C(NH)NR.sup.cR.sup.c, —C(NR.sup.a)NR.sup.cR.sup.c, —C(NO₂)R.sup.a, —C(NO₂)NR.sup.cR.sup.c, —OC(O)R.sup.d, —OC(O)OR.sup.d, —OC(O)NR.sup.cR.sup.c, —OC(NH)NR.sup.cR.sup.c, —OC(NR.sup.a)NR.sup.cR.sup.c, —[NHC(O)].sub.nR.sup.d, —[NR.sup.aC(O)].sub.nR.sup.d, —[NHC(O)].sub.nOR d, —[NR.sup.aC(O)].sub.nOR.sup.d, —[NHC(O)].sub.nNR.sup.cR.sup.c, —[NR.sup.aC(O)].sub.nNR.sup.cR.sup.c, —NHSO.sub.2R.sup.d, —[NHC(NH)].sub.nNR.sup.cR.sup.c and —[NR.sup.aC(NR.sup.a)].sub.nNR.sup.cR.sup.c; [0285] each R.sup.c is independently hydrogen, R.sup.a, or, alternatively, two R.sup.c are taken together with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of the same or different group selected from R.sup.a, =O, halogen, and R.sup.e; [0286] each R.sup.d is independently C.sub.1-6 alkyl or C.sub.3-8 cycloalkyl; and [0287] each R.sup.e is independently C.sub.1-6 alkyl, C.sub.3-8 cycloalkyl, —C(O)R.sup.d, and —C(O)OR.sup.d. [0288] 46. The pesticidal composition of clause 45, wherein the pesticide comprises an acaricide, fungicide, herbicide, insecticide, molluscicide,

nematocide, or a combination thereof. [0289] 47. A fungicidal composition, comprising [0290] a fungicide; [0291] a phytologically acceptable carrier; and [0292] a compound of formula (I):
 ##STR00144## [0293] wherein: [0294] G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R.sup.a and R.sup.b; [0295] Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C.sub.1-6 alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R.sup.a and R.sup.b; [0296] R.sup.1 is hydrogen or C.sub.1-6 alkyl; [0297] or Z and R.sup.1 together form a 5-, 6-, or 7-membered ring selected from heterocyclalkyl optionally substituted with 1, 2 or 3 groups selected from R.sup.a and R.sup.b; [0298] each R.sup.a is independently selected from the group consisting of C.sub.1-6 alkyl, C.sub.3-8 cycloalkyl, C.sub.5-C.sub.10 aryl, C.sub.6-C.sub.10 arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclalkyl, 4-11 membered heterocyclalkylalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl; [0299] each R.sup.a is optionally substituted with one or more groups selected from R.sup.b and R.sup.e; [0300] each R.sup.b is independently selected from the group consisting of =O, —OR.sup.d, —OCF.sub.3, —OCF.sub.2H=S, —SR.sup.d, =NR.sup.d, =NOR.sup.d, —NR.sup.cR.sup.c, halogen, —CF.sub.3, —CF.sub.2H, —CN, —NO.sub.2, =N.sub.2, —N.sub.3, —S(O)R.sup.d, —S(O).sub.2R.sup.d, —S(O).sub.2OR.sup.d, —S(O)NR.sup.cR.sup.c, —S(O).sub.2NR.sup.cR.sup.c, —NHS(O).sub.2R.sup.d, —OS(O)R.sup.d, —OS(O).sub.2R.sup.d, —OS(O).sub.2OR.sup.a, —OS(O).sub.2NR.sup.cR.sup.c, —C(O)R.sup.d, —C(O)OR.sup.d, —C(O)N R.sup.cR.sup.c, —C(NH)NR.sup.cR.sup.c, —C(NR.sup.a)NR.sup.cR.sup.c, —C(NO₂)R.sup.a, —C(NO₂)NR.sup.cR.sup.c, —OC(O)R.sup.d, —OC(O)OR.sup.d, —OC(O)NR.sup.cR.sup.c, —OC(NH)NR.sup.cR.sup.c, —OC(NR.sup.a)NR.sup.cR.sup.c, —[NHC(O)].sub.nR.sup.d, —[NR.sup.aC(O)].sub.nR.sup.d, —[NHC(O)].sub.nOR d, —[NR.sup.aC(O)].sub.nOR.sup.d, —[NHC(O)].sub.nNR.sup.cR.sup.c, —[NR.sup.aC(O)].sub.nNR.sup.cR.sup.c, —NHSO.sub.2R.sup.d, —[NHC(NH)].sub.nNR.sup.cR.sup.c and —[NR.sup.aC(NR.sup.a)].sub.nNR.sup.cR.sup.c; [0301] each R.sup.c is independently hydrogen, R.sup.3, or, alternatively, two R.sup.c are taken together with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of the same or different group selected from R.sup.a, =O, halogen, and R.sup.e; [0302] each R.sup.d is independently C.sub.1-6 alkyl or C.sub.3-8 cycloalkyl; [0303] each R.sup.e is independently C.sub.1-6 alkyl, C.sub.3-8 cycloalkyl, —C(O)R.sup.d, and —C(O)OR.sup.d. [0304] 48. The composition of clause 47, wherein the fungicide is selected from the group consisting of benzimidazoles, dicarboximides, phenylpyrroles, anilinopyrimidines, hydroxyanilides, carboxamides, phenyl amides, phosphonates, cinnamic acids, oxysterol binding protein inhibitors, triazole carboxamides, cymoxanil, carbamates, benzamides, demethylation inhibiting piperazines, demethylation inhibiting pyrimidines, demethylation inhibiting azoles, including imidazoles and triazoles, cyproconazole, difenoconazole, fenbuconazole, flutriafol, mefentrifluconazole, metconazole, ipconazole, prothioconazole, tebuconazole, tetraconazole, triadimefon, triadimenol, triticonazole, morpholines, cyflufenamid, metrafenone, pyriofenone, strobilurins, copper ammonium complex, copper hydroxide, copper oxide, copper oxychloride, copper sulfate, sulfur, lime sulfur, ethylenebisdithiocarbamates, aromatic hydrocarbons, phthalimides, guanidines, polyoxins, fluazinam, thiazolidines and combinations thereof.

[0305] In view of the many possible embodiments to which the principles of the disclosed invention may be applied, it should be recognized that the illustrated embodiments are only preferred examples of the invention and should not be taken as limiting the scope of the invention. Rather, the scope of the invention is defined by the following claims. We therefore claim as our invention all that comes within the scope and spirit of these claims.

Claims

1. A method for inhibiting apyrase, comprising contacting the apyrase with a compound of formula (I): ##STR00145## wherein: G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R^{sup.a} and R^{sup.b}; Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C_{sub.1-6} alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R^{sup.a} and R^{sup.b}; R^{sup.1} is hydrogen or C_{sub.1-6} alkyl; or Z and R^{sup.1} together form a 5-, 6-, or 7-membered ring selected from heterocyclalkyl optionally substituted with 1, 2 or 3 groups selected from R^{sup.a} and R^{sup.b}; each R^{sup.a} is independently selected from the group consisting of C_{sub.1-6} alkyl, C_{sub.3-8} cycloalkyl, C_{sub.5-10} aryl, C_{sub.6-10} arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclalkyl, 4-11 membered heterocyclalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl; each R^{sup.a} is optionally substituted with one or more groups selected from R^{sup.b} and R^{sup.e}; each R^{sup.b} is independently selected from the group consisting of =O, —OR^{sup.d}, —OCF_{sub.3}, —OCF_{sub.2}H=S, —SR^{sup.d}, =NR^{sup.d}, =NOR^{sup.d}, —NR^{sup.c}R^{sup.c}, halogen, —CF_{sub.3}, —CF_{sub.2}H, —CN, —NO_{sub.2}, =N_{sub.2}, —N_{sub.3}, —S(O)R^{sup.d}, —S(O)_{sub.2}R^{sup.d}, —S(O)_{sub.2}OR^{sup.d}, —S(O)NR^{sup.c}R^{sup.c}, —S(O)_{sub.2}NR^{sup.c}R^{sup.c}, —NHS(O)_{sub.2}R^{sup.d}, —OS(O)R^{sup.d}, —OS(O)_{sub.2}R^{sup.d}, —OS(O)_{sub.2}OR^{sup.d}, —OS(O)_{sub.2}NR^{sup.c}R^{sup.c}, —C(O)R^{sup.d}, —C(O)OR^{sup.d}, —C(O)NR^{sup.c}R^{sup.c}, —C(NH)NR^{sup.c}R^{sup.c}, —C(NR^{sup.a})NR^{sup.c}R^{sup.c}, —C(NO₂)R^{sup.3}, —C(NO₂)NR^{sup.c}R^{sup.c}, —OC(O)R^{sup.d}, —OC(O)OR^{sup.d}, —OC(O)NR^{sup.c}R^{sup.c}, —OC(NH)NR^{sup.c}R^{sup.c}, —OC(NR^{sup.a})NR^{sup.c}R^{sup.c}, —[NHC(O)]_{sub.n}R^{sup.d}, —[NR^{sup.a}C(O)]_{sub.n}R^{sup.d}, [NHC(O)]_{sub.n}OR^{sup.d}, —[NR^{sup.a}C(O)]_{sub.n}OR^{sup.d}, —[NHC(O)]_{sub.n}NR^{sup.c}R^{sup.c}, —[NR^{sup.a}C(O)]_{sub.n}NR^{sup.c}R^{sup.c}, —NHSO_{sub.2}R^{sup.d}, —[NHC(NH)]_{sub.n}NR^{sup.c}R^{sup.c} and —[NR^{sup.a}C(NR^{sup.a})]_{sub.n}NR^{sup.c}R^{sup.c}; each R^{sup.c} is independently hydrogen, R^{sup.a}, or, alternatively, two R^{sup.c} are taken together with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of the same or different group selected from R^{sup.a}, =O, halogen, and R^{sup.e}; each R^{sup.d} is independently C_{sub.1-6} alkyl or C_{sub.3-8} cycloalkyl; each R^{sup.e} is independently C_{sub.1-6} alkyl, C_{sub.3-8} cycloalkyl, —C(O)R^{sup.d}, and —C(O)OR^{sup.d}; provided that the compound does not have the formula ##STR00146##
2. The method of claim 1, wherein G comprises a six-membered aryl or six-membered heteroaryl ring.
3. The method of claim 2, wherein G is an optionally substituted phenyl.
4. The method of claim 3, wherein G is substituted phenyl.
5. The method of claim 4, wherein the compound has formula (Ia): ##STR00147##
6. The method of claim 4, wherein the compound has formula (Ib): ##STR00148## wherein Ar is optionally substituted aryl or heteroaryl.
7. The method of claim 6, wherein Ar in formula (Ib) is unsubstituted, substituted with an R^{sup.a} group of C_{sub.1-6} alkyl, or substituted with an R^{sup.b} group of halogen.
8. The method of claim 2, wherein G is an optionally substituted biphenyl group.
9. The method of claim 8, wherein the compound has the formula (Ib1): ##STR00149##
10. The method of claim 1, wherein G is optionally substituted pyrimidinyl.
11. The method of claim 10, wherein the compound has formula (Ic): ##STR00150## wherein Ar is optionally substituted aryl or heteroaryl.
12. The method of claim 11, wherein Ar in formula (Ic) is unsubstituted, substituted with an R^{sup.a} group of C_{sub.1-6} alkyl, or substituted with an R^{sup.b} group of halogen.

13. The method of claim 1, wherein G and its substituents have 6 to 20 non-hydrogen atoms.
14. The method of claim 1, wherein Z is substituted phenyl.
15. The method of claim 14, wherein Z is phenyl substituted with an optionally substituted aryl or heteroaryl group.
16. The method of claim 15, wherein Z is phenyl substituted with an optionally substituted alkyl group.
17. The method of claim 16, wherein Z is phenyl substituted with an R^{sup.b} group.
18. The method of claim 17, wherein the compound has a formula selected from formula (Id):
##STR00151##
19. The method of claim 17, wherein Z is phenyl substituted with an R^{sup.b} group that is halogen or —CF₃.
20. The method of claim 1, wherein Z is selected from the group consisting of optionally substituted monocyclic heteroaryl; optionally substituted pyridyl; optionally substituted pyrimidyl; and wherein Z and R^{sup.1} together form a 5-membered or 6-membered heterocyclalkyl ring.
21. The method of claim 1, wherein Z is selected from the group consisting of ##STR00152##
##STR00153##
22. The method of claim 1, wherein Z and its substituents have 6 to 20 non-hydrogen atoms.
23. The method of claim 1, wherein contacting the apyrase comprises treating a crop with the compound.
24. The method of claim 23, further comprising treating the crop with a pesticide.
25. The method of claim 24, wherein the pesticide is selected from acaricides, fungicides, herbicides, insecticides, molluscicides, nematocides, or a combination thereof.
26. A pesticidal composition, comprising a pesticide; a phytologically acceptable carrier; and a compound of formula (I): ##STR00154## wherein: G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R^{sup.a} and R^{sup.b}; Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C₁₋₆ alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R^{sup.a} and R^{sup.b}; R^{sup.1} is hydrogen or C₁₋₆ alkyl; or Z and R^{sup.1} together form a 5-, 6-, or 7-membered ring selected from heterocyclalkyl optionally substituted with 1, 2 or 3 groups selected from R^{sup.a} and R^{sup.b}; each R^{sup.a} is independently selected from the group consisting of C₁₋₆ alkyl, C₃₋₈ cycloalkyl, C₅₋₁₀ aryl, C₆₋₁₀ arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclalkyl, 4-11 membered heterocyclalkylalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl; each R^{sup.a} is optionally substituted with one or more groups selected from R^{sup.b} and R^{sup.e}; each R^{sup.b} is independently selected from the group consisting of =O, —OR^{sup.d}, —OCF₃, —OCF₂H=S, —SR^{sup.d}, =NR^{sup.d}, =NOR^{sup.d}, —NR^{sup.c}R^{sup.c}, halogen, —CF₃, —CF₂H, —CN, —NO₂, =N₂, —N₃, —S(O)R^{sup.d}, —S(O)₂R^{sup.d}, —S(O)₂OR^{sup.d}, —S(O)NR^{sup.c}R^{sup.c}, —S(O)₂NR^{sup.c}R^{sup.c}, —NHS(O)₂R^{sup.d}, —OS(O)R^{sup.d}, —OS(O)₂R^{sup.d}, —OS(O)₂OR^{sup.d}, —OS(O)₂NR^{sup.c}R^{sup.c}, —C(O)R^{sup.d}, —C(O)OR^{sup.d}, —C(O)NR^{sup.c}R^{sup.c}, —C(NH)NR^{sup.c}R^{sup.c}, —C(NR^{sup.a})NR^{sup.c}R^{sup.c}, —C(NOH)R^{sup.a}, —C(NOH)NR^{sup.c}R^{sup.c}, —OC(O)R^{sup.d}, —OC(O)OR^{sup.d}, —OC(O)NR^{sup.c}R^{sup.c}, —OC(NH)NR^{sup.c}R^{sup.c}, —OC(NR^{sup.a})NR^{sup.c}R^{sup.c}, —[NHC(O)]_nR^{sup.d}, —[NR^{sup.a}C(O)]_nR^{sup.d}, —[NHC(O)]_nOR^{sup.d}, —[NR^{sup.a}C(O)]_nOR^{sup.d}, —[NHC(O)]_nNR^{sup.c}R^{sup.c}, —[NR^{sup.a}C(O)]_nNR^{sup.c}R^{sup.c}, —NHSO₂R^{sup.d}, —[NHC(NH)]_nNR^{sup.c}R^{sup.c} and —[NR^{sup.a}C(NR^{sup.a})]_nNR^{sup.c}R^{sup.c}; each R^{sup.c} is independently hydrogen, R^{sup.a}, or, alternatively, two R^{sup.c} are taken together with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of

the same or different group selected from R.sup.a, =O, halogen, and R.sup.c; each R.sup.d is independently C.sub.1-6 alkyl or C.sub.3-8 cycloalkyl; each R.sup.e is independently C.sub.1-6 alkyl, C.sub.3-8 cycloalkyl, —C(O)R.sup.d, and —C(O)OR.sup.d.

27. The pesticidal composition of claim 26, wherein the pesticide comprises an acaricide, fungicide, herbicide, insecticide, molluscicide, nematocide, or a combination thereof.

28. The pesticidal composition of claim 26, wherein the pesticide comprises an herbicide.

29. A fungicidal composition, comprising a fungicide; a phytologically acceptable carrier; and a compound of formula (I): ##STR00155## wherein: G is monocyclic aryl or monocyclic heteroaryl; wherein G is optionally substituted with 1, 2, or 3 groups selected from R.sup.a and R.sup.b; Z is selected from the group consisting of monocyclic aryl, bicyclic aryl, monocyclic heteroaryl, bicyclic heteroaryl, and C.sub.1-6 alkyl; wherein Z is optionally substituted with 1, 2 or 3 groups selected from R.sup.a and R.sup.b; R.sup.1 is hydrogen or C.sub.1-6 alkyl; or Z and R.sup.1 together form a 5-, 6-, or 7-membered ring selected from heterocyclalkyl optionally substituted with 1, 2 or 3 groups selected from R.sup.a and R.sup.b; each R.sup.a is independently selected from the group consisting of C.sub.1-6 alkyl, C.sub.3-8 cycloalkyl, C.sub.5-C.sub.10 aryl, C.sub.6-C.sub.10 arylalkyl, 2-6 membered heteroalkyl, 3-8 membered heterocyclalkyl, 4-11 membered heterocyclalkylalkyl, 5-10 membered heteroaryl and 6-16 membered heteroarylalkyl; each R.sup.a is optionally substituted with one or more groups selected from R.sup.b and R.sup.e; each R.sup.b is independently selected from the group consisting of =O, —OR.sup.d, —OCF.sub.3, —OCF.sub.2H=S, —SR.sup.d, =NR.sup.d, =NOR.sup.d, —NR.sup.cR.sup.c, halogen, —CF.sub.3, —CF.sub.2H, —CN, —NO.sub.2, =N.sub.2, —N.sub.3, —S(O)R.sup.d, —S(O).sub.2R.sup.d, —S(O).sub.2OR.sup.d, —S(O)NR.sup.cR.sup.c, —S(O).sub.2NR.sup.cR.sup.c, —NHS(O).sub.2R.sup.d, —OS(O)R.sup.d, —OS(O).sub.2R.sup.d, —OS(O).sub.2OR.sup.d, —OS(O).sub.2NR.sup.cR.sup.c, —C(O)R.sup.d, —C(O)OR.sup.d, —C(O)NR.sup.cR.sup.c, —C(NH)NR.sup.cR.sup.c, —C(NR.sup.a)NR.sup.cR.sup.c, —C(NOH)R.sup.a, —C(NOH)NR.sup.cR.sup.c, —OC(O)R.sup.d, —OC(O)OR.sup.d, —OC(O)NR.sup.cR.sup.c, —OC(NH)NR.sup.cR.sup.c, —OC(NR.sup.a)NR.sup.cR.sup.c, —[NHC(O)].sub.nR.sup.d, —[NR.sup.aC(O)].sub.nR.sup.d, —[NHC(O)].sub.nOR.sup.d, —[NR.sup.aC(O)].sub.nOR.sup.d, —[NHC(O)].sub.nNR.sup.cR.sup.c, —[NR.sup.aC(O)].sub.nNR.sup.cR.sup.c, —NHSO.sub.2R.sup.d, —[NHC(NH)].sub.nNR.sup.cR.sup.c and —[NR.sup.aC(NR.sup.a)].sub.nNR.sup.cR.sup.c; each R.sup.c is independently hydrogen, R.sup.a, or, alternatively, two R.sup.c are taken together with the nitrogen atom to which they are bonded to form a 5 to 8-membered heterocyclalkyl which may optionally include one or more of the same or different additional heteroatoms and which may optionally be substituted with one or more of the same or different group selected from R.sup.a, =O, halogen, and R.sup.e; each R.sup.d is independently C.sub.1-6 alkyl or C.sub.3-8 cycloalkyl; and each R.sup.e is independently C.sub.1-6 alkyl, C.sub.3-8 cycloalkyl, —C(O)R.sup.d, and —C(O)OR.sup.d.

30. The composition of claim 29, wherein the fungicide is selected from the group consisting of benzimidazoles, dicarboximides, phenylpyrroles, anilinopyrimidines, hydroxyanilides, carboxamides, phenyl amides, phosphonates, cinnamic acids, oxysterol binding protein inhibitors, triazole carboxamides, cymoxanil, carbamates, benzamides, demethylation inhibiting piperazines, demethylation inhibiting pyrimidines, demethylation inhibiting azoles, including imidazoles and triazoles, cyproconazole, difenoconazole, fenbuconazole, flutriafol, mefentrifluconazole, metconazole, ipconazole, prothioconazole, tebuconazole, tetraconazole, triadimefon, triadimenol, triticonazole, morpholines, cyflufenamid, metrafenone, pyriofenone, strobilurins, copper ammonium complex, copper hydroxide, copper oxide, copper oxychloride, copper sulfate, sulfur, lime sulfur, ethylenebisdithiocarbamates, aromatic hydrocarbons, phthalimides, guanidines, polyoxins, fluazinam, thiazolidines and combinations thereof.
